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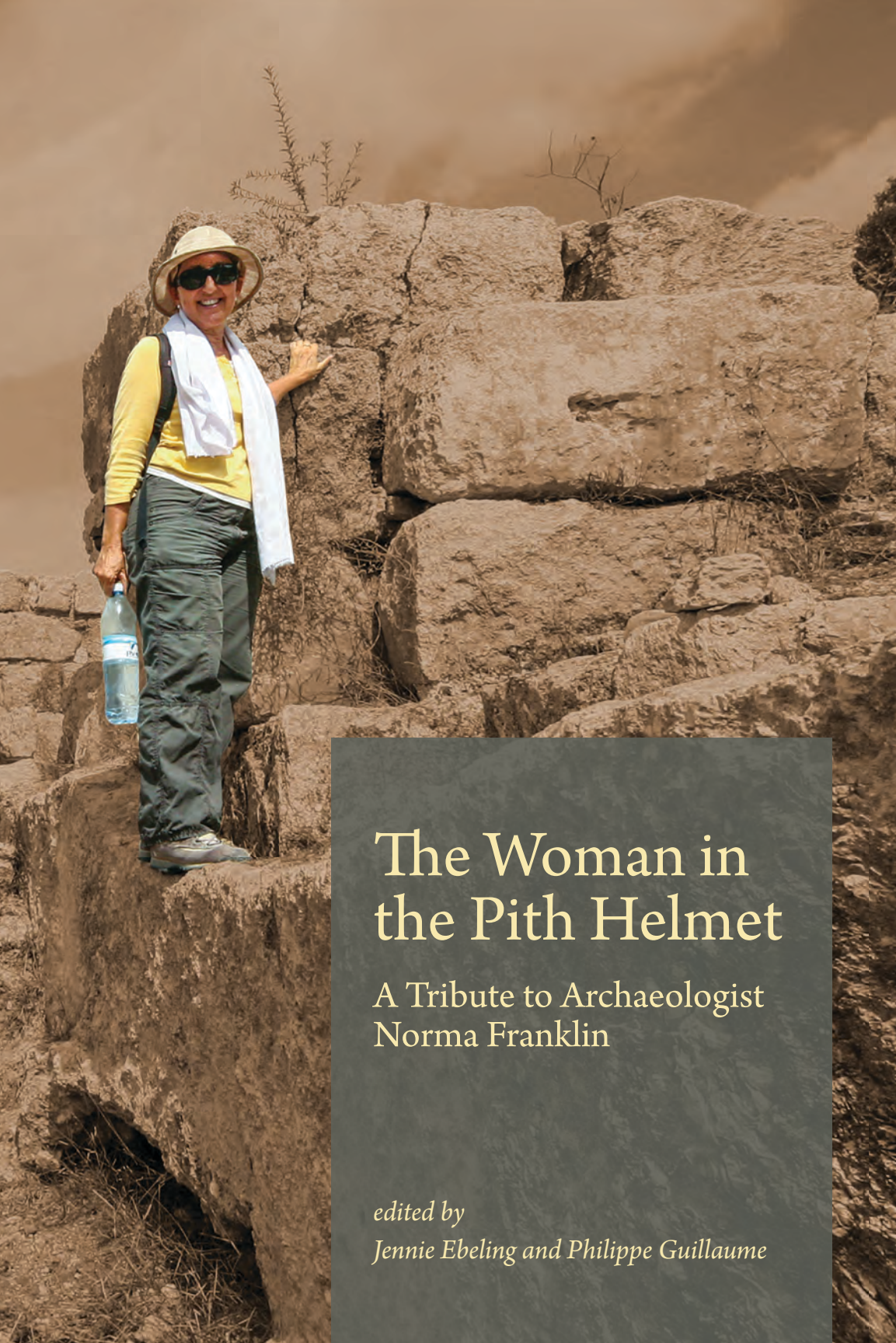
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The Woman in the Pith Helmet

A Tribute to Archaeologist
Norma Franklin

*edited by
Jennie Ebeling and Philippe Guillaume*

THE WOMAN IN THE PITH HELMET

A TRIBUTE TO ARCHAEOLOGIST NORMA FRANKLIN

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LOCKWOOD PRESS

2020

THE WOMAN IN THE PITH HELMET

A TRIBUTE TO ARCHAEOLOGIST NORMA FRANKLIN

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Excavating Jezreel

Martha Hellander

Like reapers at harvest,
We advance across the golden field.
Wielding pickaxes and *tourias*
We attack the wild oat-straw,
Slash the yellow-petaled mustard,
Flay the tough-stemmed fennel;
Scrape topsoil, pry stones,
Evict gerbils and scorpions that flee our blows,
To lay out squares at *Tel Jezreel*.

Points marked and string untangled,
Loci assigned and buckets tagged,
We unspool the wide black cloth,
Tie canopies to poles with cryptic knots
For life-giving shade:
Our *Beyt-haYaar-haLevanon*.
A sigh of morning breeze
Brushes our bare arms;
Long-buried ones sense our presence and stir,
As the sun rises over *Ein Jezreel*.

We lay siege to the shrine,
Five centimeters at a time.
We tell tales, interpret finds:
This limestone boulder shaped by an adze—
A foundation stone of Ahab's palace!
These rock-cut limestone pits—
The winery of the *tzaddick* Naboth!

I come seeking Jezebel.
Daughter of Tyre, garden city on an island,
Sent to marry Ahab, king in dusty Israel,
To seal a covenant with family ties,
Far from her beloved sea.
Dispatched in a coup at Jezreel,
Or so the story goes.

Jezebel stands at her window,
Adorned in Hathor headdress,
With sacred mask of malachite and kohl,
Her necklace the Tears of Isis.
The people pass below,
Mourning the young god whose death
Is reenacted each spring—
Communal grief, in which all have a share,
Gives way to joy at resurrection—
Jezebel leading the dance.

Jezebel as priestess hears the grieving citizens,
Jezebel as goddess blesses their sacred tears.
When her killer drives up,
Chariot wheels smoking for full effect,
Jezebel as Queen Mother calls out *murderer*
And stares him down.

Horseshoe nails dug from the road to the tell suggest Jehu's horses.
Their sharp hooves dismembered Jezebel's body,
Devoured by Ashera's sacred dogs,
Leaving only the sacrificial parts:
Her skull, her hands, her feet,
Or so the story goes.

At the bottom of a cistern,
The skeleton of a dog.

Might this cleft be Jezebel's tomb,
Divined by porcupine quills,
With amethyst scarab and dedicated jug?
Might that hollowed shard be her skull?
Narrow flints honed from rubbing glaze on pots

Evoked her finger bones, delicate and long,
Tracing lines of text as she reads the law
In the royal archives of Tyre.

Long before Jezebel's time,
Basalt was shaped and smoothed here,
Into standing stones and offering bowls,
Grinding stones and basins.
Holiness imbues this place
Above the sacred spring.
The Gilboa, an arrow's flight away,
Lies like a recumbent goddess
Who disdained holy war
And gathered peace-loving kings
Into her body, the first temple.

The bowls, the standing stones, the pots
Topple and shatter in the holy wars
In the blood-soaked Valley of Jezreel.
They bear the weight of war-horses.
Our voices awaken them.
We lift the relics, hold them aloft.
They enjoy a brief moment in the sun,
And begin to tell their secret Things:
How the people hulled and threshed the emmer,
Made beer, fermented mash for the dead;
How incense smoldered in the stone vessel;
How the grapes were borne aloft and trampled,
Yielding wine for the great festivals
Of harvest and thanksgiving.
An ancient *baetyl*,
Its slender curve of finely-honed basalt
Set upright after millennia underground,
With the warm skin of a mother held close,
Summons unexpected tears.

We share among ourselves
Words of trees and whispers of stone,
Glimpses of holiness
That abided here,
Long before the gods had names.

By mid-day our shades flap and strain, saying
Leave this stony outcrop to bake
A few more centuries, let the basalt
Crumble and cleave in the burning sun.
The shades long to join the black storks
That soar over the *tel*,
In the hot wind from *Beth She'an*.

On the dig, discipline prevails.
Artifacts are handed in,
Measured, weighed,
Labeled, drawn,
Photographed and wrapped,
Packed into crates dark inside
As the earth they came from.
The shades are neatly folded.
The pickaxes, the *tourias*,
The buckets, the survey tools,
Decommissioned, yet ever imbued
With the dust of Jezreel.

QUMRAN IN THE IRON AGE, WITH CROSS-TEMPORAL REFLECTIONS ON THE HASMONEAN AND EARLY ROMAN PERIODS

JOAN E. TAYLOR AND SHIMON GIBSON

Abstract: While the site of Qumran has been of great interest in connection with the period relating to the depositing of the Dead Sea Scrolls in nearby caves, with the later periods of occupation at the site heavily prioritized, the earlier Iron Age II period of occupation (seventh–sixth centuries BCE) has been relatively neglected. In this paper, the existing evidence for the Iron Age II settlement of Qumran is dealt with thoroughly. The authors propose that in this period Qumran consisted of a rectangular enclosed settlement with a central courtyard, a separate domestic unit, a round cistern and water system, and an industrial area with two pottery kilns. It was clearly not an insignificant cluster of huts, as has recently been suggested. It is demonstrated that it belonged to a string of settlements along the western coast of the Dead Sea. Understanding the structural layout of the Iron Age site is vital because the template of the site was largely maintained in the subsequent Hasmonean resettlement.

Keywords: Qumran, Roland de Vaux, Buqei'a plateau, 'Ein Feshkha, Iron Age Judah

From 2005 onward, the present authors have been engaged in understanding Qumran within its wider landscape, particularly in regard to the paths connecting Qumran to other localities around the Dead Sea and inland to the Buqei'a, the Judaeen wilderness that stretches from the top of the Qumran pass to Hyrcania and beyond.¹

The conclusions we have reached in this study are important for Qumran, in that we were able to determine that the pathway systems around the site were fundamentally Iron Age in terms of their basic template and character.² These paths—including

1. This article is dedicated with affection to Dr. Norma Franklin, an indefatigable and perceptive researcher of the Iron Age in the southern Levant, and a good friend of one of the authors of this paper (Gibson). Norma joined us, together with Louise Hitchcock, for an unforgettable visit with a picnic in January 2009 to view the Iron Age sites investigated by the late Professor Lawrence E. Stager in the Buqei'a plateau above Qumran to the west.

2. Taylor and Gibson 2008, 2011. To further the research for this article, the authors made a number of visits to Qumran and environs in April and August 2004, February 2005, November 2010, May 2011, June 2012, and August 2014, to clarify various architectural and stratigraphical details, and



Figure 1. A map of the northern end of the Dead Sea showing sites mentioned in the text. Map prepared by J. E. Taylor.

the pathways of the pass—were more or less maintained for local traffic in later periods, but there was never any significant further development, meaning that the old paths

to visit ongoing excavations. We are grateful to the Palestine Exploration Fund for financial support, also to the late Yuval Peleg for showing us the results of his excavations at the site with Yitzhak Magen, and to Randall Price and Oren Gutfield for showing us their digging operations in the area of the southern esplanade. A draft of this paper was delivered to the International SBL conference in London in 2011, and we are also grateful to the audience at this event for comments received, especially to Stephen J. Pfann, Jodi Magness, and Dennis Mizzi.

sufficed for the limited use made of them during the ages that followed. In researching the “road system” around Qumran, our work was continually pulled away from the well-known settlement of Qumran in the time of the Dead Sea Scrolls, to another time hundreds of years earlier (the Iron Age II) when this region, including the hinterland of the Judean wilderness, was seemingly much more developed.

After exploring the Iron Age II at Qumran with a view to understanding the road system more holistically, our study has led us to understand Qumran itself as lying at a crossroads in the Iron Age, situated on the main route along the Dead Sea coast, where settlements were dotted between Jericho and ‘Ein Gedi, and located at the junction that led to the pass and the Judean wilderness settlements (fig. 1). Researchers who have seen Qumran as being on this important crossroads in the Herodian and Early Roman periods have made a fundamental error of dating. The connectivity of Qumran was far more apparent in the Iron Age than at any other time, just as these settlements were more numerous and significant in the Iron Age: in the Hasmonean and Herodian periods, none of the settlements of the Buqei’a Plateau existed.

Here we will not focus on the material objects discovered from the Iron Age³ but rather consider the architecture of the site on the basis of the remnants of walls and reports of excavations. It will be argued that the extent of the Iron Age at Qumran has been considerably underestimated by previous scholars, and that some features previously attributed to the late Hellenistic/Hasmonean period, particularly from Period Ia, are more likely to be of Iron Age origin.

The Iron Age II at Qumran has been attributed to the late eighth century to the early sixth century BCE based on pottery finds—a period of about one hundred years or so—and so it is likely that this period might itself be divided into different architectural phases.⁴ At present, however, we will focus on the main delineations of walls and other features, and consider their significance.

3. We are very grateful to Mariusz Burdajewicz and Jean-Baptiste Humbert who kindly permitted us to read the report on the Iron Age pottery intended for the final Qumran publication. Burdajewicz notes that there are 350 diagnostic pieces from the period that he dates from the late eighth to the early part of the sixth century BCE from both the site of Qumran and the caves. The pottery comprises basic common ware well known at Judahite sites from the Iron Age IIC: jars, jugs, cooking pots, large bowls/kraters, smaller bowls, plates, and lamps, but the assemblage did not include any interregional imported wares or luxury vessels. See now the suggested second half of the seventh century to early sixth century BCE date for the Iron Age assemblage at Qumran, and its Judahite “uniformity,” in Yezerski 2018. Yezerski illustrates seventy-seven items in her publication, but it is unclear what percentage this is of the overall Iron Age assemblage that was examined from the site. Vessel types are the same as those studied by Burdajewicz, except for a few basins, a stand, a lid, a tripod-bowl, and flasks. Yezerski also notes the particular importance in terms of quantity of the different kinds of bowls evident in the Iron Age Qumran assemblage.

4. This last phase of the Iron Age from the seventh and sixth centuries BCE is generally referred to in publications as the Iron Age IIC and now more frequently as the Iron Age III. See Gitin 2015.

The History of Qumran Excavations and the Iron Age

When Qumran excavations in the 1950s were jointly undertaken by the Palestine Archaeological Museum, the Department of Antiquities of Jordan, and the École Biblique et Archéologique Française de Jérusalem, headed by Fr. Roland de Vaux of the École Biblique⁵ (figs. 2–4), it was determined that the first constructions at the site were from Iron Age II (referred to as “israélite” in de Vaux 1956: 535–37; 1973: 1–3). A significant amount of fragmentary Iron Age pottery was discovered dated to the eighth (at the earliest) to early sixth centuries BCE (see above, n. 3). In his synthesis of results, presented as the Schweich Lectures and published in 1973, de Vaux concluded that a rectangular building with a large open courtyard and residential quarters existed on the eastern side of the site, with a water system on the west (fig. 5).

In his field notes, further details were supplied (Humbert and Chambon 1994). De Vaux noted that some wall foundations dating to the first century BCE appeared to be at a lower level than others. These lower walls originated in the Iron Age since they relate to a layer of ash with ceramic finds dating from that time (de Vaux 1973: 1). This layer was noted at the angle of the southeast corner between L.73 (Humbert and Chambon 1994: 315) and L.80, and extended further south to L.68. It was found also in the area outside the wall of L.6 (and L.40), in a zone later designated L.145 (Humbert and Chambon 1994: 338).

At the south of L.66, de Vaux observed that the later water channel running to cistern L.71 cuts through a large wall 0.90 m wide, probably dating to the Iron Age (Humbert and Chambon 1994: 314, photos 174 and 176). The deep ashy deposit was found north of L.38 and throughout this area, where the layer is associated with earlier walls, against the north wall and under the south wall of L.77, and in the room L.86–L.87/89 (Humbert and Chambon 1994: 316, 320). Since de Vaux did not explore beneath many of the floor levels of Period I, it is striking that when he did so he invariably recorded Iron II remains, though the picture is patchy. The Iron Age was clearly not his interest: in L.125 de Vaux indicated some disappointment when after opening a small trench, it yielded nothing except Iron Age sherds (Humbert and Chambon 1994: 332).

The foundations of the wall running north–south along the east of cistern L.117 were identified as belonging to the Iron Age along with other walls associated with L.106, L.109, L.113, L.114, L.119b, and L.108, with an opening to the north between L.114 and L.119, through which surface rainwater from the surrounding area was drained into a large round cistern L.110 (de Vaux 1973: 2).

5. Preliminary reports: de Vaux 1949, 1953, 1954, 1956, 1959, 1961, 1973. See also Laperrousaz 1966, 1976, 1980. We express reservations regarding some of his interpretations. The first final report of de Vaux's excavations, including significant reinterpretations, has now appeared in Humbert, Chambon, and Młynarczyk 2016. For the purposes of this paper we will continue using de Vaux's locus numbers.



Figure 2. Father de Vaux at Qumran in 1966. Photograph courtesy of Gibson Picture Archive.



Figure 3. General view of Qumran in the 1950s to the northeast. Note the tents of de Vaux's expedition next to the site. Photograph courtesy of Gibson Picture Archive.

In addition, de Vaux identified the long wall running south along the southern esplanade as Iron Age, suggesting that it served to demarcate this region. It is of the same orientation as the eastern side of the Iron Age enclosure, and the type of construction is



Figure 4. Qumran at the time of de Vaux's excavations in 1953, from L.49/L.50 to the west. Photograph courtesy of Gibson Picture Archive.

similar to the northern wall of this enclosure. The pottery identified by de Vaux did not seem to predate the seventh century BCE. He noted, however, that in L.68 there was a jar handle with a stamped seal impression in Palaeo-Hebrew characters, "to the king" (*lmlk*), of apparent late eighth century BCE date, a point we will return to presently (de Vaux 1973: 3, 59–60).

De Vaux suggested that the Iron Age settlement was eventually destroyed and burnt in the sixth century BCE; the layer of ash was noted as having associated sherds from that time. De Vaux's view was that this settlement was devastated during the Babylonian invasion of Judaea and the sacking of Jerusalem in 587/586 BCE (de Vaux 1973: 91–93), but it might have occurred slightly later in 582 BCE at the time of Nebuchadnezzar's campaign to Ammon and Moab in Transjordan (Herr 1999).

De Vaux's plan of the Iron Age remains at Qumran, published as plate III in his Schweich Lectures volume, is quite instructive concerning the locations at the site where he thought Iron Age walls must have existed. In graphic terms, the published version of de Vaux's plan is not very clear, and so we have added key locus numbers, and the heavy hatching of the reconstructed lines of walls has been eliminated (fig. 5). It should be compared with our new plan of Iron Age Qumran (fig. 7),⁶ and this shows

6. Our plan is an accurate reconstruction of the layout of Iron Age Qumran, showing the position of the architectural remains, based on the wall remnants as reported by the excavators and on early wall foundations seen by different observers (including ourselves) below later Period Ib architecture, and with the addition of the findspots where Iron Age fills, floors, and artifacts were found. But we

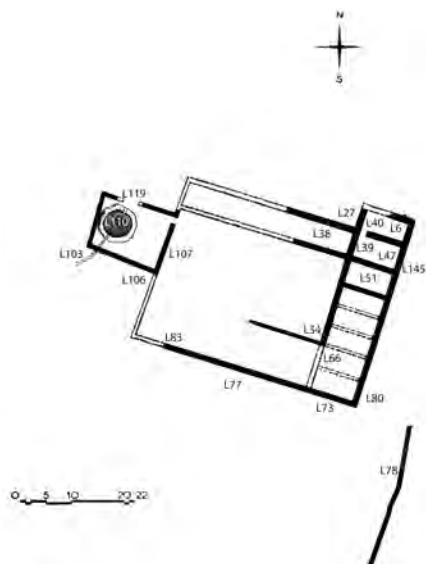


Figure 5. Plan of Qumran in the Iron Age as published by de Vaux, with the addition of locus numbers and the clarification of the appearance of the reconstructed walls. Plan adapted with modifications from de Vaux 1973: pl. III.

in red the architectural features we think existed in the Iron Age and in yellow stratigraphic locations where Iron Age levels were encountered. While these attributions cannot be absolutely certain, these will be explained during the course of this discussion.

Clearly, de Vaux's plan does not represent the exact position of the Iron Age wall fragments as he found them; instead it shows the general alignment of walls based on his field observations (shown in fig. 5 in black), and hypothetical reconstructed lines of walls (shown in white with broken lines). What emerges is that de Vaux was absolutely certain of the Iron Age date for the entire outer perimeter wall of the compound seen on the east (from L.6 and L.145 to L.80) and south (L.73 to L.83), or at least for their lower foundations (now largely buried). Based on the shape of the three rooms he identified at the northeast corner

of the compound (L.40 and L.6; L.39 and L.47; L.51), de Vaux thought there may have been five additional rooms of similar size extending to the south along the interior eastern perimeter wall.

In addition to what was previously published of de Vaux's excavations, there has now been a substantial reinterpretation undertaken by Jean-Baptiste Humbert, along with detailed plans, drawings and photographs from the archives of the de Vaux excavations (Humbert, Chambon, and Mlynarczyk 2016). This allows for some further testing of some of de Vaux's conclusions, and clarifications of his field notes. Along with this, archival photographs of de Vaux's excavations, taken by John Allegro in 1954, have enabled us to further understand the site, as have our own observations.⁷

must emphasize that it is still not an excavation plan and therefore it incorporates a certain amount of reasoned and well-considered conjecture. Thus it should be taken only as a useful working guide to the original appearance of Iron Age Qumran.

7. The photographs come from the Manchester Museum Allegro Collection, and have been digitized as part of the Dispersed Qumran Caves Artefacts and Archival Sources (DQCAAS) project, funded by the Leverhulme Trust. They are reproduced here with the permission of the copyright holder Judith Brown (© Allegro Estate), Allegro's daughter.



Figure 6. A sounding made in the 1950s revealing the Iron Age ashy layer beneath the floor at the western end of L.77. Photograph by John Allegro, © Allegro Estate, courtesy of Manchester Museum/Judith Brown.

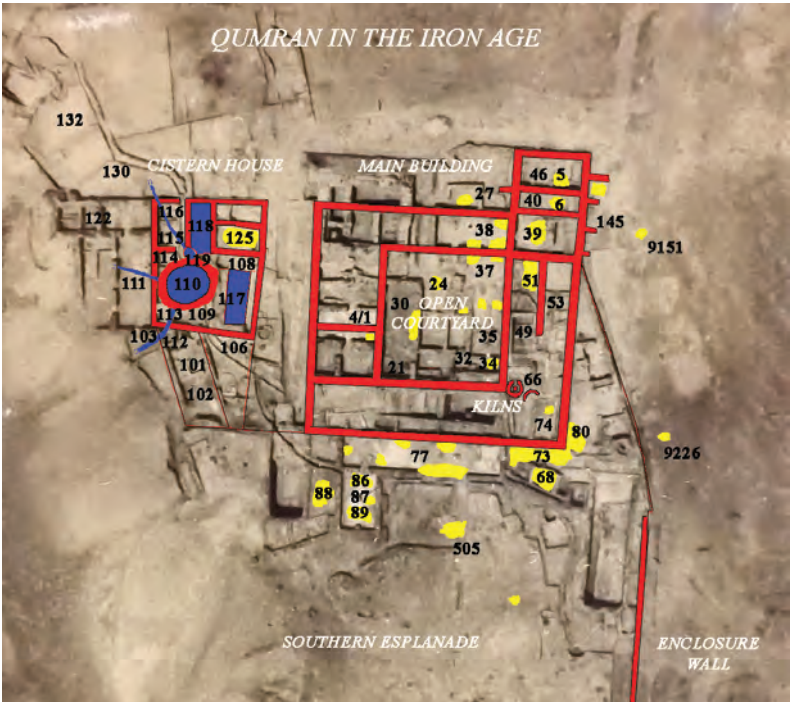


Figure 7. Plan of Qumran in the Iron Age, as suggested by the authors, on the basis of an aerial photograph published in de Vaux 1973: pl. II. Map prepared by S. Gibson.

For example, on the basis of his field notes, it appears that de Vaux's working hypothesis was that the large south wall of L.66 was from Period I and the principal channel from Period II; but, at the time of his synthesis, he changed the latter period to Period Ib and the former to the Iron Age, continuing into Period Ia. This conclusion is preferable, because the wall met the eastern wall of the Iron Age enclosure at the south-east corner of L.74, where, as we have seen, de Vaux mentioned there was evidence of ash and sherds dating to the Iron Age. In his field notes he mentions this ash in relation to L.73, commenting that this lay against cistern L.68 very deeply, being around 0.50 m thick there, while it was less thick in other places. This ash contained many (150) sherds from Iron Age II, though without complete forms. The ashy layer passes under the exterior drain of the bath of L.68. An ostrakon with two lines of Palaeo-Hebrew was discovered at this point in L.68 (KhQ 1236).⁸ In Humbert's reexamination, de Vaux's helpful stratigraphical drawing of a cut through L.68 to L.72 is reproduced, along with photographs illustrating the ashy layer.⁹ Humbert suggests that the ash, which lies thicker toward L.72–73, is the result of the cleaning out of the kilns into what is designated as L.178 (lower L.66; see below and Humbert, Chambon, and Mlynarczyk 2016: 243). Humbert notes that this Iron Age deposit of ash appears to lie against the bath of L.68 and is also found in the older basin in L.72, thus the bath of L.68 was dug into this ashy layer and the ash was flung into the former basin of L.72 (Humbert, Chambon, and Mlynarczyk 2016: 243).

In regard to the Iron Age ashy layer of L.77, a photograph by John Allegro from 1954 illustrates the sounding de Vaux made was on the west side of L.77 against wall 850. De Vaux noted in his diary that the sounding, in the vicinity of the stone circle, showed that there was under the plaster (of the upper floor) a deeper place, then some ashes and finally the virgin earth. He observed that they collected Iron II sherds in the ash layer, and that it extended under the south wall of L.77¹⁰ (fig. 6). Humbert has understood this layer to be part of the spread of ash from the kiln installation,¹¹ and

8. This was analyzed by André Lemaire and published in the second volume of the Qumran excavations, with a reading indicating the letters: ויין (line 1) and רשיים (line 2), the first line reading "and wines" and the second we suggest may be "poor" (plural): Lemaire 2003: 353, and see PAM 42–683. The second line is read differently by Lemaire, as רשיים [פ], "two horse-riders."

9. See Humbert, Chambon and Mlynarczyk 2016: fig. 92c, 92d, 94d and see the drawing on p. 249. Humbert notes (p. 243) that more accurately it is not so much directly about L.68 and L.73 but the junction of L.72 and L.73. It is not a cut across L.68 and L.73, but exterior to the former and cutting through the latter.

10. De Vaux wrote on 23 March 1954: "Un sondage autour du cercle de pierres montre qu'il y a sous le plâtre [du sol] enduit plus profond, puis des cendres et enfin la terre vierge. Dans les cendres, on ne recueille que quelques tessons du Fer II. La couche de cendres passe sous le mur sud de 77" (Humbert and Chambon 1994: 316).

11. Humbert, Chambon, and Mlynarczyk 2016: 321, though he also observes the ashy layer continued into L.86 (329, 342).

suggested that the cut was in the middle of L.77, due south of the kiln, but Allegro's photograph now shows it was beside the western wall (850).

In his field notes for L.5 de Vaux recorded the appearance of a wide wall running east–west over L.5 and L.46 with not only potsherds but also the rim of an Iron Age II hole-mouth jar (Humbert and Chambon 1994: 233), found by the southern part of the wall (Humbert and Chambon 1994: 27, photo 25). This probably indicates that the thick wall (272) is the northern edge of the Iron Age enclosure. A photograph by John Allegro from 1954 from L.19 looking east appears to show an earlier foundation wall below the Period Ib wall (274) on the west side of room L.46. (fig. 9) In addition, beyond this area to the east, in what is now identified as L.145, there were at least three small walls jutting eastwards at right angles from the main wall beyond L.6 and L.47, but these were not shown on de Vaux's plan, and, in the publication of the field notes and photographs by Humbert and Chambon, these walls appear only in the plan of Period Ib (plates IV and VIII). They do now appear also in Humbert's new analysis, labeled walls 396, 397 and 398 (Humbert, Chambon, and Mlynarczyk 2016: pl. I). These may perhaps be identified as small lean-to outbuildings adjacent to the northern room in L.5 and L.46, and further south, dating to the Iron Age.

Humbert notes regarding L.145 that the presence of Iron Age sherds witnesses to a level from this time (Humbert, Chambon, and Mlynarczyk 2016: 229–30). He notes also that de Vaux tended to refer to excavations in this area as “east of 44” or “exterior to 44” (even “east of 45” or “84”) without specificity, but de Vaux's archives have shown that he cut a trench exterior to wall 900 (running north–south) beyond L.84 in 1953 and continued this work in 1954. In this trench de Vaux observed an earthquake fissure running north–south, and also the lower ashy level associated with the end of the Iron Age site lying on virgin soil (feature 145.4). The ashy level sinks down into the fissure, but continues eastward to the edge of the trench.

Publication of the drawings of these zones enables the relevant walls to be determined more precisely. The plans drawn by Chambon in the latest Qumran volume are very helpful in determining Iron Age wall remains attested by de Vaux. Evidence of the thick north–south perimeter wall (272, 284, 285) and other associated walls of this period are sometimes shown as looking stony (not colored in black), where they differ from later walls. As mentioned in his synthesis, further west in L.38, at its south-east angle, de Vaux uncovered part of a north–south wall (267) and an east–west wall (262) with associated ash and potsherds from the Iron Age (Humbert and Chambon 1994: 305, photo 41). In the excavation notes de Vaux wrote that the wall between L.35 and 49 (292), like that between 35 and 34 (306), descends deeper than the excavated floor, and that these walls were perhaps “Israelite,” but the material was missing to date them.¹² De Vaux's comments here do not always relate clearly to published photo-

12. Humbert and Chambon 1994: 304. “le mur entre les loci 35 et 49, comme celui entre 35 et 34,



Figure 8. The wall of the esplanade near L.78 towards the west. Photograph from the 1960s, courtesy of Gibson Picture Archive.



Figure 9. View from L.14 toward L.27 in a photograph from the 1950s. Note the Iron Age stone foundation at the base of the wall. Photograph by John Allegro, © Allegro Estate, courtesy of Manchester Museum/Judith Brown.



Figure 10. The kiln at L.66. Photograph de Vaux 1973: pl. Vb.

graphs, but his observations in L.38 and L.34–L.35 formed the basis for his reconstruction of an inner Iron Age wall running parallel to the outer perimeter wall.

De Vaux also noted traces of an east–west Iron Age wall running to the southwest corner of L.34, between L.32 and L.36, continuing the southern wall of L.36 (295) (Humbert and Chambon 1994: 304). A lower foundation, which is probably a western continuation of this wall, is evident at L.21 at the southern end of L.30 (240).

The evidence of de Vaux's surviving exposed walls suggests that the eastern perimeter wall ran from the northeast corner of L.6 to the southeast corner of L.74, a distance of 37 m. The question of the north, south, and west walls of the Iron Age enclosure are, however, not clarified by de Vaux's notes, or by the reexaminations by Humbert. This is because the lowest level of the site was not substantially explored. In general, de Vaux did not dismantle walls from Period I in order to define the extent of the Iron Age settlement beneath it, preferring instead to focus on the periods related to the Dead Sea Scrolls. He made only occasional soundings, but these are quite instructive (as we shall see below).

Regarding the Iron Age date for the potters' kilns, close to the inner wall of the Iron Age enclosure de Vaux identified two kilns (L.66), one quite destroyed but the

descend plus bas que le sol dégagé. Murs peut-être israélites mais le matériel manque pour le dater." The intersection of these two walls may be shown in photos 76 and 77, but this is unclear.

other in a relatively good state (fig. 10). These had been covered over by a stone pavement following changes in the ground level owing to the creation of the later cistern L.48–49 (Humbert and Chambon 1994: 314). De Vaux wondered in his field notes whether these may be dated to Iron II, but by the time of his synthetic publication (De Vaux 1973) he had changed his mind, and assigned these instead to Period Ia. During this period, which de Vaux dated to Qumran's resettlement in the late second century BCE, a similar enclosure to the one from the Iron Age was identified, but with certain additional features. Despite the destruction of the Iron Age settlement, de Vaux concluded that "what remained of the Israelite buildings served as a point of departure for the constructions of the new occupants" (de Vaux 1973: 4).

In L.66 de Vaux indicated that the two kilns, built side by side, were obliterated during the development of the site in the next period (Ib), when a large cistern was built to the north (L.48–49, de Vaux 1973: 16–17). He stated simply that "there is nothing to indicate that these kilns were already in service during the Israelite [=Iron Age] period" (de Vaux 1973: 4). This is curious as we have seen that he identified the inner wall very near the kilns at the southeast corner of L.34 as Iron Age, meaning that their alignment and positioning fits well with these. De Vaux did not record that there was any layer of Iron II ash and sherds there. Instead, he associated pottery from Period Ib from the southern area of the main building with these kilns. The levels he refers to are unclear. The absence of a record of Iron II pottery fragments does not in fact mean there were none, since de Vaux did not undertake pottery seriation during his sorting procedures, and many of the supposedly nondiagnostic body sherds he may have thrown away. In general, one must be wary of some of the stratigraphical associations provided by de Vaux, owing to the sparse details provided in his field notes.

Jodi Magness has pointed out, on the basis of the evidence de Vaux provided for Period Ia as a whole, that "de Vaux found no coins associated with Period Ia, and there were only a few potsherds which he could not distinguish in type from those of Period Ib" (Magness 2002: 64). This has led her to assign most of the supposed Period Ia developments to Period Ib, beginning in the first quarter of the first century BCE.¹³ However, importantly, Magness noted that the kilns in L.66 are more appropriately seen as Iron Age, since they "represent a kind of circular kiln with a central pillar that is attested in Palestine from the Bronze Age on" (Magness 2002: 64). This seems the most plausible solution, and one also now adopted by Humbert (see above), and we therefore assign the kilns to Iron Age II, allowing us to view the southern corner of the main Iron Age enclosure as a pottery workshop.

As noted, it was south of L.66, near L.73, that the jar handle with the *lmlk* ("for the king") stamped seal impression was found (KhQ1235).¹⁴ Another of these was found in the excavations of Magen and Peleg in their L.3000 next to the enclosure wall in

13. For a good summary of Period Ia, see Mizzi 2015.

14. Humbert and Chambon 1994: 314; Lemaire 2003: 353; see PAM 42-683. De Vaux 1973: 2

the southern esplanade (see below). It has formerly been accepted that the particular type of stamp in KhQ1235 should be dated to 705–701 BCE, to the time of the revolt of King Hezekiah against the Assyrian king Sennacherib (Ussishkin 1977). However, Oded Lipschits has proposed a new chronological scheme, based on a typology set out by André Lemaire (1981). This type of jar handle is classified by Lemaire as type XII, with a two-winged sun and the inscription *lmlk* appearing above the upper part of the solar disk, with nothing written in the lower part (Lemaire 2003: 353). This is one of the jar handle types that Lipschits has noted appear only in hill-country sites that were not destroyed in 701 BCE, or in strata of the seventh century BCE, one of the “late types” (Lipschits, Sergi, and Koch 2010: 11, 13–17; Lipschits 2012: 1, 8). This would provide evidence consistent with the view that the settlement survived through to the Babylonian invasion of 587/586 or 582 BCE.

We concur with de Vaux in considering that the end of the Iron Age II occupation of Qumran is represented by a destruction layer of ash. Where ash appears at the site, therefore, it is important to review whether it might belong to this level.¹⁵ For example, following the removal of the floor in L.30 de Vaux encountered an oven, which he attributed to Period Ia. At a lower level was a layer of ash that appeared to extend beneath the eastern wall, but he does not indicate the date of the pottery from within this layer. Is this also to be dated to the Iron Age? Along the western wall of L.132 it is clear that in Period Ib a wall that begins in L.142 has been destroyed, leaving no remains (Humbert and Chambon 1994: Period Ib, pl. IV). Water erosion led to the destruction of the western wall of L.132. The preserved wall built in Period II and buttressing on the northwestern corner of the western building was a rebuild from the ground up, with buttressing on the southwest of L.130 that went down to a thin layer of ash (Humbert and Chambon 1994: 333). De Vaux thought that this might be from a fire sometime at or after the end of Period Ib, but is it possible that this is the Iron Age destruction level ash which appears in a confused picture at the corner of this building as a result of the flooding and rebuild, resulting in a partially reversed stratigraphy?

To seek further answers concerning the stratigraphy of Qumran, renewed excavations at the site of Qumran took place between 1994 and 2004, undertaken by Yitzhak Magen and Yuval Peleg.¹⁶ The excavations focused on four “refuse” dumps, to the north, south, and east of the main building, and, in addition, a large paved area immediately

cites it as L.68, but de Vaux must have been confusing it with the other Iron Age sherd with Hebrew letters (KhQ1236).

15. For a discussion of the question of the ash and carbonized materials at the site, see Humbert, Chambon, and Mlynarczyk 2016: 47–55. Humbert links Iron Age II ash in the south with the kilns (p. 50), but it seems too substantial and spread out over the site to be the result of dispersed ash from one locality (L.66 area).

16. Yitzhak and Peleg 2006, 2007, 2018. The authors are grateful to the late Yuval Peleg for showing us around the site in 2004, pointing out to us the exact places where they had uncovered Iron Age II remains, and discussing with us their possible relevance.

to the south of L.77 was exposed. Test squares were also made inside the buildings, all of which revealed the same deep layer of ash and small sherds previously noted by de Vaux (Magen and Peleg 2018: 101–4; 2007: 27). An area on the southern plateau was also excavated (Magen and Peleg 2007: 4, fig. 4).

The southern refuse dump of the later settlement was excavated, located south of pool L.71 and extending west to the paved area south of L.77, and Iron Age pottery was found mixed with Period Ib pottery (L.401 Magen and Peleg 2018: 59). In the paved area south of L.77 Iron Age pottery was found as residuals beneath the pavement (e.g., L.505: Magen and Peleg 2018: fig. 51). In addition, Iron Age II pottery was found in the northern refuse dump, some 25 m north of the site at the edge of the plateau (L.2028), and in an area immediately east of the main building (L.9151, 9226).¹⁷ In the northern part of this area, an intact Iron Age juglet was found (Magen and Peleg 2018: 68–69; 2007: 28; 2006: 76).

During their work inside the main building, Magen and Peleg were able to identify a number of locations with Iron Age remains and pottery.¹⁸ A hint as to the extent of the Iron Age on the west side of the main building was obtained as a result of a sounding Magen and Peleg excavated beneath the eastern side of the floor of room L.4, revealing the top of an Iron Age wall.¹⁹ This wall runs from west to east, where it evidently bonded with another wall with a north to south axis, probably running beneath wall WS19, and therefore delimiting the western side of the courtyard. Yet, another early wall which may also be from this period was visible on the interior side of wall WS29 at the southern end of room L.30 (Magen and Peleg 2018: cross-section 1-1 in fig. 25). This wall may very well be the western extension of an Iron Age wall running to the west from the direction of L.32 and L.34 (see above). Magen and Peleg assume that the original shape of the central courtyard was determined no earlier than Period Ib, but this flies in the face of the evidence of their own soundings made in 1998 (L.3038, 3045, 3046, 3062, and 3063) which revealed much Iron Age pottery in this area. More-

17. Magen and Peleg 2018: 59–60 (eastern refuse dump), 68–69 (northern refuse dump); 2007: 6–10; 2006: 59–64.

18. The exact stratigraphical association between the pottery published by Yezereski and the Iron Age walls, floors and fills at the site is unclear. The locus list in the final report is not particularly helpful in that regard and does not include chronological data or pottery readings, which of course reduces its usefulness: Magen and Peleg 2018: 141–69. The following might be helpful to future researchers: Yezereski published seventy-seven items, out of which eleven came from the main building (L.3038, 3045, 3046, 3062, 3063), three from the western wing of the main building (L.9148, 9154, 9164), one from the southeast angle of the main building (L.9160), six from the eastern refuse dump (L.9151, 9226), three from the south side of the building in L.77 (L.9521), three from the paved square (L.505, 3006, 3022), thirty-nine from the southeast side of the site (L.507, 1001, 1002, 1003, 3000), two from the southern esplanade (L.353, 401), seven from the northern refuse dump (L.508, 510, 522, 801, 2023, 2028), and four from unclear locations (L.421, 603, 1101).

19. Magen and Peleg 2018: 23: L.4/1, figs. 22–24, 27. The pottery found in the fill adjacent to this wall was not included in the final report.

over, these soundings confirmed that the walls at the northeast angle of the courtyard at L.37 had to be from the Iron Age (Magen and Peleg 2018: 48). New work took place on the southeast side of the building, unearthing the same Iron Age layer previously seen by de Vaux in L.73, L.80, and under Pool L.68. As noted above, this is where de Vaux found a *mlk* stamped jar handle. The new excavations revealed Iron Age sherds and stone weights (L.3033, 9106, 9148; Magen and Peleg 2018: 103).

In room L.6 within the northeast angle of the main building, the removal of a floor revealed a wall with a west to east axis (W575) and an ashy layer; Magen and Peleg suggested dating it to the Iron Age (Magen and Peleg 2018: figs. 31–32; 103). On the other side of the east wall of the building, in the area of L.145, Magen and Peleg recorded three or four “poorly built” walls dating from the Iron Age (Magen and Peleg 2018: 59). In the eastern wing, in the area de Vaux previously designated as L.51, Magen and Peleg found the foundation of a wall (W600) with a north–south axis, which they attributed to the Iron Age (L.9148, 9154, 9164; Magen and Peleg 2018: 103, figs. 41, 43). A southern extension of this same wall was traced in L.53 and was seen running beneath W589, which is the eastern wall of the stepped installation L.48/L.49. At least three underground stone-lined pits identified as silos (L.341/1–3) were found in the southern esplanade (L.353), but their Iron Age date is not absolutely certain.²⁰ Segments of walls made of fieldstones were unearthed, jutting out from the eastern boundary wall of the esplanade (W576); the associated layer was described by the excavators as “close to and under the eastern outer wall” (L.1005, 3000, 3004). Beneath this ashy layer was an earthen and cobbled living surface with a number of hearths (Magen and Peleg 2018: 103). The Iron Age layer of ash and sherds appeared to lie beneath the long esplanade wall (as de Vaux found under the southern wall of L.77), and here, abutting the wall on the southeastern side, another *mlk* seal impression on a jar handle was found (L.3000; Magen and Peleg 2018: 104; 2007: 27, figs. 29–31; 2006: fig. 3:24).

Further excavations by Randall Price and Oren Gutfield in the area of the southern esplanade uncovered Iron Age pottery, but this has not yet been published.²¹ At any rate, this area to the south of the settlement would undoubtedly have been in use during the Iron Age. The esplanade is flat and elongated, with a length of approximately 160 m from north to south, and a breadth of 60 m, narrowing considerably to 12 m in the south. It may have been bounded by a wall on the eastern side. The history of the construction of this eastern esplanade boundary wall is still not fully understood. It would appear to have served as an architectural terrace supporting and retaining the

20. Magen and Peleg 2018: 69 mentioning four pits, figs. 79–81, and p. 102 mentioning three pits, with the authors adding: “based on the sherds found between the wall’s stones, they date to the Iron Age”; 2007: 32, fig. 32; 2006: 36, 38: fig. 3.25. We note that these silos have the same appearance as the circular pits containing Early Roman pottery which have been excavated elsewhere on the southern esplanade.

21. We are grateful to the excavators Randall Price and Oren Gutfield for graciously showing us the results of their work in the southern esplanade on a number of occasions between 2004 and 2012.



Figure 11. The esplanade wall to the west, with the late Yuval Peleg on the right in 2004. Photograph by S. Gibson.



Figure 12. The esplanade wall running to the south, as it appeared in 2004. Photograph by S. Gibson.

ground surface to its west; clearly it was rebuilt on a number of occasions²² (fig. 11). A recent examination suggests that it had to be contemporary with the earliest Iron Age use of the esplanade; otherwise the ground surface within this area would have eroded away (fig. 12). The layer of ash and Iron Age sherds noted by Magen and Peleg probably extends within a breach in a destroyed or collapsed segment of the original wall, which was subsequently rebuilt in the Hasmonean period. The narrow walls jutting out from this esplanade wall may represent an earlier phase of construction within the Iron Age, or perhaps a “casemate” arrangement of rooms or magazines connected to the esplanade wall (as at sites in the Buqei’a and at nearby Khirbet Irneh, see below).²³ The fact that this terrace wall was in use during the entire existence of the Qumran settlement down to the first century CE is not at all surprising for architectural terracing, reflecting a continuous use over a long period.²⁴

Excavations within the main building uncovered another north–south wall inside L.51 and L.53, continuing east of pool L.48 and pool L.50. This seems to continue the line of a later dividing wall between L.39 and L.47 (fig. 13), which would indicate that later constructions were indeed founded on Iron Age foundations. However, Magen and Peleg conclude from their investigations of the eastern wall, attributed by de Vaux to the Iron Age, that “the said foundations are an integral part of the Hasmonean structure.” Nevertheless, in L.6 today it seems that the outer eastern wall is built over the ruins of the Iron Age outer wall, which is used as a foundation (fig. 14). Magen and Peleg, however, assume that it is only the small walls jutting out at right angles from the line of the eastern boundary, and under the long wall of the southern esplanade, that represent the true Iron Age structures, so that the settlement from that period “was composed of clay huts and wooden sheds that rested, in part, on fieldstone building foundations” (Magen and Peleg 2018: 11, 104; 2007: 28). This is curious given the very large amount of Iron Age sherds and ash found throughout the site, and a small settlement of simple huts does not account for the thick wall of L.51, though here they suggest that “life in the settlement possibly revolved around a public structure or stone tower” (Magen and Peleg 2018: 104; 2007: 28). Moreover, a collection of small huts is not a form paralleled in other Iron Age II settlements in the region (Faust 2003, 2018).

Further to the south inside the main building a stone weight calibrated with lead was found; this appears to have been discovered in some proximity to the kilns of L.66 (Magen and Peleg 2006: fig. 3:26). A large amount of ash and sherds of Iron Age II date was found under the Period I floor at the west end of L.77 on top of a lower plaster floor

22. Humbert has suggested the extension of the wall south to ‘Ein Feshkha as an *eruv*, but this relates to the Hasmonean period; Humbert, Chambon, and Młynarczyk 2016: 35–45.

23. Our use of the term “casemate” refers to a series of very small rooms running along one side of a long wall, and it does not imply they had any defensive function; Lapp 1976.

24. Indeed, the longevity of this wall as a functioning terrace is quite remarkable and in 1856 it was the most prominent visible feature at the site when visited by one of the early explorers; Isaacs 1857: 66 engraving opposite.



Figure 13. The architectural features at L.51, toward the west. Photograph by S. Gibson.



Figure 14. The east wall of the building complex, just south of L.145, viewed toward the west. Photograph by J. E. Taylor.

(L.9520, 9522 and 9525; Magen and Peleg 2018: 45–46; 2007: 27). This is interesting given de Vaux’s observation of this layer under the southern wall (Magen and Peleg’s W613), suggesting that there was another southern wall of this Iron Age structure to the south of L.77 (i.e., under the “paved square”) that has yet to be found. This may

have been an extension of an initial enclosure, since the basic template of an Iron Age site like that of Qumran was generally rectangular, as we see at such nearby single-era sites as Hurvat Shilhah (Kh. Shilheh; Mazar, Amit, and Ilan 1995). This site offers a good comparable structure to the form of Iron Age Qumran.

On the western side of the site, de Vaux's round pool L.110 (fig. 15) was clearly shown not to stand alone. Magen and Peleg were able to determine that the original height of the round pool L.110 was about 1 m below its later form, and it had a close relationship with L.117 and L.118 from the beginning, suggesting to them that both these pools were used for gathering clay in their sediment (fig. 16; Magen and Peleg 2018: 50–57, figs. 57–58; 2007: 13).

Magen and Peleg are doubtful that the round cistern could be Iron Age, believing that the hydraulic plaster is typical of the Hellenistic and Hasmonean periods.²⁵ Nevertheless, as Yizhar Hirschfeld suggested, the Iron Age cistern might simply have been replastered in order to make it waterproof and useful in later times (Hirschfeld 2004: 57). Indeed, its extended height over which the coating of plaster was attached clearly indicates that some of the plaster dates to the time of this later modification.²⁶ Magen and Peleg note that they did not find Iron II remains on the western side of the site, and thus they assumed this area was used for collecting rainwater solely in the Hasmonean period (Magen and Peleg 2018: 103–4; 2007: 27). But this very observation seems to suggest that (a) a cistern complex was indeed required in association with any rainwater collection, which would have been necessary also in the case of the Iron Age, and (b) that rainwater collection, involving occasional flooding, could have washed away considerable primary evidence from this lower level. In actual fact, de Vaux did uncover Iron Age remains in L.125, which lies within the cistern complex, immediately to the northeast. It remains to be seen whether further soundings will uncover Iron Age remains there.

In Magen and Peleg's view, the cisterns cannot be Iron Age because, as they state:

It defies belief that the Iron Age inhabitants of Qumran, who themselves lived in huts, were capable of digging such a huge pool ... the powerful character of Pool L-110 is inconsistent with that of the site's Iron Age structures. It is very likely that the Iron Age pools were quite small, and probably dried up before the end of summer. In fact, Iron Age Qumran was quite small, and may well have been inhabited only in winter and spring. (Magen and Peleg 2007: 28)

25. Unfortunately, we still lack good comparative material on plaster types. Impressive first steps have been made by Yosef (Sefi) Porath and Aryeh Shimron in their research, but a greater database of plaster types is essential to make this a viable dating method. In our opinion, the earliest plaster coating near the base of L.110 could very well be Iron Age.

26. This is further complicated by the fact that modern restoration appears to have been made to the plaster coating the interior of this installation, at least in its upper parts.



Figure 15. The round cistern L.110 as seen in 2010. Photograph by S. Gibson.



Figure 16. The cistern house toward the west, with the round cistern L.110 center left, and the stepped installation L.117 on the left, in a photograph taken in 1977. Photograph courtesy of Gibson Picture Archive.

This observation seems to fly in the face of their own excavations that brought to light a large amount of Iron Age artifacts, especially potsherds. The suggestion that the long wall bounding the southern esplanade was not constructed in the Iron Age but in a later period, over the conflagration layer and Iron Age walls, seems unlikely owing to its terrace-like function in supporting the ground surface within the esplanade. This does not, however, negate the possibility of “casemate” rooms built into or adjoining the long wall. Moreover the template for Iron Age (and Persian) settlements usually involved rectangular enclosure structures, with “casemate” rooms or magazines along the perimeter and with an open central area, though there could be outlying buildings at times and animal pens surrounded by stone fences (Faust 2003, 2018). At Qumran, it is possible there were structures abutting the main enclosure, and also various buildings in the southern esplanade.

The evidence of thick Iron Age walls uncovered by de Vaux and forming the basis for a main rectangular Iron Age enclosure still seems clear from his field notes, plan, and photographs. Magen and Peleg’s new thick wall of L.51 also fits into this template, as does their evidence of a plastered floor lying underneath L.77, indicating a continuation of the Iron Age presence to the south.

Rather than an area of small huts, we suggest an activity zone with a rectangular compound with a central courtyard, with a possible domestic structure at the northeast angle, and fenced-off areas of Iron Age II date, with possible building extensions to the east and south. Magen and Peleg argue that since Pool L.110 had to have been built at the same time as Cistern L.117 and L.118, which they dated to no earlier than the Hasmonean period, this meant that an Iron Age date for Pool L.110 was untenable. In addition to this they claimed that the Iron Age inhabitants at the site would in any case have been incapable of constructing a large pool with thick hydraulic plaster (Magen and Peleg 2007: 28). In our opinion, the exact opposite can be argued, and both circular reservoirs and stepped cisterns of Iron Age date are paralleled in other sites.²⁷ The round cistern with the adjoining stepped cisterns L.117 and L.118 and its surrounding walls, would then be lying on the west side of this enclosure, but not necessarily an integral part of the main building, as de Vaux had suggested. In addition, given a possible minimum of seventy years of occupation in the Iron Age II, there need not have been one single timespan of construction after which the site was left exactly in the same state. Indeed, it is more likely that there were successive developments within this time frame, with at least a couple of phases of building construction.

In regard to the notion that the Iron Age settlers could not build a sophisticated water system, it is apparent throughout the Kingdom of Judah in urban and industrial contexts, and in the neighboring region of the Buqei’a where the art of water harvesting in fields was extremely developed. We contend that an advanced water system would

27. Tsuk 2001–2002; Miller 1980. On the Iron Age water system at Suba with a stepped cistern, see Gibson 2009.

have been imperative for the pottery works at the site in the Iron Age. Apart from kilns, a pottery workshop would have required storage and stacking surfaces, clay preparation installations, and of course an ample source of water. Hence, the Qumran system must have included the round pool (L.110) for the collection of water and the stepped cisterns (L.117 and L.118) for the preparation of the clay, and all this would be appropriate given the presence of the Iron Age pottery kilns within the main building (L.66).²⁸ If the clay preparation area was on the west where there was access to water, with the kilns inside the building to the east, then the drying floors, where vessels were stacked until leather-hard and ready to be fired, might have been in the intervening courtyard, or alternatively toward the southern esplanade (perhaps the plastered floor under L.77 was one such working surface).

In his synthesis of the chronology of the site, de Vaux concluded that in Period Ia new occupants expanded the water collection and storage system in the western area around the old round cistern. A channel was built to supply water more effectively to the round cistern and two other rectangular cisterns built close by, L.117 and L.118, with a decantation or silting basin (L.119) serving all three cisterns. It is now possible to determine, on the basis of Magen and Peleg's excavations, that this in fact describes the form of the original construction of the cistern complex. In addition, de Vaux (1973: 3–4) noted that a small channel was seen running underneath an Iron Age wall, in order to collect rain water from the south, and a covered building was constructed in the south (L.101–102), with other rooms added to the north (L.115–116 and 125–127).

From his field notes it seems that an early wall lay along the north of L.106 and L.101 (L.112) and was cut into by the main water channel developed in Period Ib (Humbert and Chambon 1994: 329). Another lower and narrow water channel was first encountered in L.103, to the west, running from L.112. In L.122 de Vaux dug to determine the continuation of this but found it disappeared in L.112. But it seemed possibly linked with another channel, on virgin soil, running from L.111, which he found turning to L.113. On lifting a wall between L.109 and L.113, he found the first lower floor, coated with plaster, which communicated by a narrow opening with the supply channel to L.110. In L.109 on lifting the lower threshold between L.109 and L.101, another lower floor was encountered anterior to the principal channel and without doubt contemporary with the first circular cistern (Humbert and Chambon 1994: 326). In L.119 de Vaux was clearly fascinated in tracing the progress of the lower channel systems, and in his field notes he concluded that water was drained from a roof above L.101 and L.106, and that therefore the three cisterns L.110, L.117 and L.118 “sont contemporaines” (Humbert and Chambon 1994: 329). Strangely, he decided against his own field observations in his later synthesis. However, Magen and Peleg's new excavations now affirm the integral connections between all three installations, as de Vaux

28. For the various installations associated with a clay-production plant from the Iron Age, see Gibson 2009.

himself had observed, and they have also uncovered the earlier system of channels, with an entrance channel in the north wall of L.116, running beneath the floor of L.116 and L.115, ultimately ending up in the silting basin L.119 (Magen and Peleg 2006: 66).

To sum up, we suggest the original layout of the rectangular building complex surrounding Pool L.110 and Cisterns L.117 and L.118, is from the Iron Age and it should not be attributed to the elusive (and now apparently nonexistent) Period Ia, as de Vaux assumed. Rainwater was channeled into the installations from the surroundings, as well as collected from the roofs of the rooms round about. This covered cistern-house system may conceivably be similar to the *beth ha-bor* mentioned in Jeremiah (37:15–16, 38:6). A complex water system would not be surprising for this period and numerous water-collection and storage installations, many exhibiting a high degree of technological and engineering skills, have hitherto been investigated at both urban, village, and countryside locations, as for example at Beersheba, the Amman Citadel, Beth Shemesh, Suba, and elsewhere (Faust 2008c; Haiman 2002).

This sophisticated water installation fell into disuse at the time of the destruction of the settlement in the sixth century BCE. It was subsequently restored and put back into use when the site was redeveloped at the beginning of Period I, ca. 100 BCE. When the aqueduct was built, new covered water channels were made and large pools of this era were also constructed, and the old installations were put back into use. The major overhaul and restoration of this installation complex in Period Ib, has ultimately confused the overall stratigraphic picture of the western side of the Iron Age settlement, despite the fact that the architectural evidence clearly indicates the cisterns must predate the Hasmonean development of the site.

We therefore offer a new plan of Qumran in the Iron Age. De Vaux's suggestion that there may have been five additional rooms of similar size extending to the south along the interior eastern perimeter wall is unlikely owing to the discovery by Magen and Peleg of a wall with a north to south axis, running at the center of de Vaux's room L.51, and also because we suggest assigning the kilns in L.66 to the Iron Age (see below), which makes the existence of de Vaux's two southern rooms unlikely. Our reconstruction of the walls at the northeast corner of the compound differs from that of de Vaux, and it is based on the alignment and visual appearance of the underlying walls in this area. Moreover, the wall stubs at L.145, and possibly also at L.27, suggest an additional northern room must have existed at the northeast angle enclosing L.5 and L.46. What emerges is that the compound had a separate and integrated building at the northeast angle with at least two distinct rooms; it is likely to have been used for dwelling purposes and perhaps even for basic defensive purposes as well. Long corridors ran along the interior of the northern (L.38), eastern (L.55 to L.66), western (L.4/1), and southern (L.74 to L.83) perimeter walls; some of these may have been subdivided into rooms for storage. These corridors surrounded an open rectangular courtyard at the center of the compound (extending from L.37 on the north, to L.34 at the inner southeast angle, and to L.30 on the west).

De Vaux also attributed the earliest use of the cistern house (L.110), on the western side of the compound, to the Iron Age, and indicated that it was surrounded by four walls (L.106, L.107), with a possible doorway at L.119 (shown reconstructed on his plan), and a water channel entering the cistern from L.103. Our plan of the cistern house includes the installations L.117 and L.118, and additional rooms to the north. De Vaux identified the retaining wall on the eastern side of the southern esplanade as Iron Age and traced it as far north as L.78 (fig. 8), but he was uncertain as to its northern continuation and where it connected with the earlier compound.

The Identification of Iron Age Qumran

Iron Age Qumran has featured very little in the comprehensive discussions dealing with the archaeology of the site, but it has been of interest to scholars of the Hebrew Bible, especially in regard to discussions on the identification of the toponyms mentioned in Josh 15. Even prior to the discovery of the Scrolls, there was some speculation that the ruins of Qumran might represent the remains of a biblical site. F. M. Abel (1938: 386) identified Qumran with Middin. Martin Noth (1938: 72; also Cross 1956: 12–17) identified the site as Ir ha-Melah, the “City of Salt,” mentioned in Josh 15:61–62: “In the wilderness: Beth-Araba, Middin and Secacah and Nibshan and the City of Salt and Engedi; six cities with their villages.”

De Vaux followed Martin Noth’s identification, including the suggestion that the other Judaeen settlements of the Judaeen monarchy were sequentially indicated in this passage.²⁹ Pessah Bar Adon likewise identified Ain el-Ghuweir as the City of Salt and Qumran with biblical Secacah (Josh 15:61; Bar Adon 1977). This suggestion is followed also by Jodi Magness (2002: 24–25). Hanan Eshel (2009: 8) has also agreed that the site was Secacah (or Sokokah in the LXX), stating its name is “likely from the root s-k-k (‘to cover’), referring to the date-palm leaves used to cover the roofs of the houses.” John H. Allegro (1959–1960: 64) suggested a relationship between Qumran and the name of the Kippah torrent mentioned in the Copper Scroll, near which there is a place called Sekaka (3Q15 IV, 13; V, 2, 5, 13).

However, while the general town list of Josh 15, does approximate the geographical scope of the small kingdom of Josiah (639–609 BCE), one must be cautious (Ahituv 1993). The identification of the Iron Age sites of the Judaeen wilderness, which include ‘Ein Gedi, would need to be on the basis of substantial “cities” that could have smaller “villages” in their environments. Since there are many more Iron Age sites known now than those indicated by the named localities in Josh 15:61–62, we would need first to determine where the larger “cities” might have been, apart from Jericho and ‘Ein Gedi.

29. See de Vaux 1973: 91–92; Cross and Milik 1956. The LXX deviates from the Masoretic Text here, and not only in the forms of the names, but there is no reference to “in the wilderness.” Most likely, these names indicate other actual sites known at the time of the composition of the LXX, at which time there were no settlements in the wilderness.

The area of the wilderness would need to encompass the region of the higher hinterland (the Buqe'ā) and also the coastal region of the Dead Sea. Altogether this is a wide area, and unknown sites may yet come to light.

Iron Age Context: Eastern Judah in the Eighth–Sixth Centuries BCE

De Vaux pointed out that in 2 Chr 26:10, King Uzziah (781–740 BCE) is described as having large herds and loving the soil, that he “built towers” (*migdalim*) in the wilderness and hewed out many cisterns. Prior to this, Jehoshaphat (870–848 BCE) was said to have built in Judah fortresses and store cities, but there are no archaeological indications of building efforts in the wilderness (2 Chr 17:12; de Vaux 1956). One could argue therefore that the earliest structures in the wilderness should be Uzziah’s “towers” constructed in the eighth century BCE. However, the archaeological picture we encounter is entirely different: apart from large settlements such as Jericho (Tell es-Sultan), with settlement remains extending from as early as the ninth and eighth centuries BCE, the landscape on the eastern edge of Judah consists of a plethora of small sites, all thriving during the seventh century BCE, but only a fraction of them have pottery finds extending back to the late eighth century BCE.

As Magen and Peleg note (2007: 25, fig. 28; 2006: 73–79, fig. 3.22), the establishment of the settlement of Qumran in the Iron Age was part of a great wave of settlement in this area in the late eighth and early seventh centuries. Their map indicates the variety of known categories of Iron Age sites in the wider region, with solitary towers, cave-dwellings, fortified enclosures, industrial locations, and town sites. They suggest that settlement of this region was not a result of a specific state strategy relating to the initiatives of individual Judahite kings, but was the direct result of the settlement of refugees following the fall of Samaria in 720 BCE; these refugees are mentioned in Jer 41:5 and 1 Chr 30:10–11; 34:9. Avraham Faust, however, has argued that Judah was at its settlement peak in the seventh century BCE, especially in what he calls the “fringe areas” of the Judaeian Desert and the Negev, despite the catastrophe of Sennacherib’s invasion of 701 BCE. During the seventh century BCE this region “suddenly witnessed an unparalleled wave of settlement activity” (Faust 2008b: 170; Stern 1994: 406–7).

In terms of the main Iron Age sites in the region which were contemporary to the settlement of Qumran, there is of course the city of Jericho (Tell es-Sultan), which was in continuous occupation from the eighth through sixth centuries BCE, though it had an earlier occupation in the Iron Age as well.³⁰ Further south is Kh. el-Uja (Bar Adon 1977: 109, no. 26), Ard al-Mafjar (Bar Adon 1977: 113–14 site no. 59), and Tell es-Samarat (Bar Adon 1977: 114, site no. 61). Following this, there is Vered Yericho, and

30. On Kathleen Kenyon’s work on the Iron Age II at Jericho (Tell es-Sultan), see Kenyon and Holland 1982. For recent work on the third phase of the Iron Age II stratum at Jericho (“Sultan VIc,” dated 732–535 BCE), see Nigro 2014. For a survey of this area see Bar Adon 1972.

another Iron Age site without a name, 3 km south of Tell es-Sultan, followed by a large site of 30 dunams 3 km north of Vered Jericho and in Wadi Qelt (Eitan 1983; Horowitz, Tchernov, and Lernau 2018). Further south is Kh. Muhalhil (Bar Adon 1977: 114, 117–33). To the east, on the northern side of the Dead Sea, Rujm el-Bahr is a small stony island which served as an artificial anchorage (Bar Adon 1989; Hirschfeld 2006: 221–22). Various Iron Age II sherds were found by Bar Adon at the site but recent researchers believe these to be intrusive, i.e., they were brought to the site as residuals during later construction activities.³¹ At Wadi Mukallik (Nahal Og) south of Kibbutz Almog, there is a cluster of three Iron Age II sites, and finally we reach Qumran.

At the top of the rocky pass to the west of Qumran is the Buqei'a plateau and on it are several Iron Age II sites, with the main ones situated at Kh. el-Maqari, Kh. es-Samrah, Kh. Karm Abu Tabaq, Kh. Abu Tabaqi, as well as two further unnamed sites; all of these were explored by Frank Cross and Józef Milik in the 1950s, and later by Lawrence E. Stager in the 1970s (fig. 17; Stager 1975, 1976; Cross and Milik 1956: 7; Cross 1993; Master 2009; Ilan 1973: 186–94). These Iron Age sites had sophisticated water harnessing systems, particular in the case of Kh. Abu Tabaq, Kh. es-Samrah, and Kh. al-Maqari, which also had large open reservoirs.³² The landscapes immediately surrounding these sites were divided up into fenced plots of land with well-defined irrigation systems, dams, and retaining walls, now also partly destroyed. The appearance of these sites and the pottery found there suggested to Cross and Milik that the agricultural farming at these Buqei'a sites was a centrally organized activity conducted by the ruling authorities at the time of the reigns of Hezekiah or Josiah. What was grown in the fields on the plateau is unknown,³³ but we suggest they were sown primarily to provide green pasturage and irrigated using flood-farming techniques. Grazing would have been in the late winter and spring, with herds relying on stored fodder, dry stubble and natural shrubs for the rest of the year, with some transhumance occurring as well toward higher rainfall areas in the summer months. We suggest therefore that the settlements on the plateau were used as protected havens for semi-sedentary specialized pastoralists, with corrals for livestock, storage facilities, and access to water, probably used on a seasonal basis, during the seventh–sixth centuries BCE. While there are evident ecological constraints in the Buqei'a, with limited and unpredictable water sources and a lack of arable soil and other resources, aspects typical of a marginal environment, the archaeological evidence actually suggests otherwise and the region clearly thrived in the Iron Age II.³⁴

31. On the new work at Rujm el-Bahr see Oren et al. 2015.

32. The Iron Age sites on the Buqei'a plateau were examined by the authors in October 2007, together with Rafael Y. Lewis and Mareike Grosser, as part of our roads investigation funded by the PEF. Another visit was made in January 2009 by Gibson, together with Norma Franklin and Louise Hitchcock.

33. Stager 1976: 146 suggested the cultivation of cereals and legumes.

34. On other marginal environments that were not so marginal, see Berger and Juengst 2017; Legge 2004.



Figure 17. An aerial view of the Buqei'a plateau toward the north. Photograph by S. Gibson.

Just south of Qumran, less than 1 km away, is an area with a few scattered ruins, which appears on Jack Ziegler's map under the name "Airneh" (fig. 18). It is here that de Vaux identified and partly excavated an Iron Age II structure, with a set of "casemate" rooms at the southwest corner of a large square enclosure (60×64 m) with two square "towers" at the southeast and northeast corners respectively (Humbert and Chambon 1994: 367, pl. XLVIII). It was never fully published, and it is now in a very bad state of preservation, with the modern road going through the eastern part of the structure. We reexamined the site, remeasuring the structural remains, and tentatively naming it Khirbet Irneh³⁵ (fig. 19). Our reexamination revealed the entire western boundary wall (1 m thick) belonging to the structure (62 m in length), with two separate ranges of "casemate" rooms along the inner side, at the southwest angle (23×3.50 m) and at the northwest angle (16.5×3.50 m). A gap (4 m wide) in the wall may have been the entrance into the compound (No. 5). The southwest range of rooms (Nos. 1–4) was excavated by de Vaux, consisting of four narrow and small rooms (4.80×1.50 m; 2.50×2.00 m; 4.70×2.00 m; 2.00×1.50 m), one which had a stone-cobbled floor (No. 1). We traced the northwest range of rooms (No. 6), not seen by de Vaux, and one can

35. The site was visited by the authors in October 2007 and November 2010, and measurements were taken with the assistance of Rafael Y. Lewis and Mareike Grosser.

detect the outlines of at least two small rooms (No. 7–8), but they remain unexcavated. The northern boundary wall (0.50 m thick) was traced for 19 m, and the southern wall for 28 m. It had pottery that was dated by de Vaux to the eighth century, suggesting it was superseded by the building on the plateau.³⁶ The surface Iron Age pottery we examined, however, was not particularly diagnostic, and it could very well be from the seventh century BCE.³⁷ The overall size of the enclosed compound (0.37 ha) suggests it was used primarily as a corral to contain livestock.³⁸

Nearby, to the east, was another isolated Iron Age structure (12 × 12 m) which was linked to a zig-zagging Iron Age stone wall, running close to the dried-up spring of 'Ein Ghazal, which was also explored by de Vaux³⁹ (fig. 20). This wall and small building appear to have been destroyed in modern agricultural clearance work, and our repeated attempts to locate the original position of these remains have been unsuccessful.

The lower wall associated with the Israelite fortress extending toward 'Ein Feshkha was most likely built to delineate the plantation region irrigated by small springs. It resembled the construction method of the southern esplanade wall at Qumran, even though the latter was a terrace not a stone fence. In addition, there were various walls and small stone structures between Irneh and 'Ein Feshkha.⁴⁰ At 'Ein Feshkha, Yizhar Hirschfeld (2004: 171, 184) excavated an Iron Age structure on the hill to the west of the present road.

A late Iron Age fortified settlement existed at Kh. Mazin (Qasr el-Yahoud), south of 'Ein Feshkha,⁴¹ but recent work shows the anchorage there is of later date because sea levels in the Iron Age were quite low (ca. 405 to 408 mbsl; Oren et al. 2015: 90). At 'Ein el-Ghuweir a structure was excavated dated to the eighth century BCE (Bar Adon 1989), and south of the site there is a long wall with rooms and houses, with structures running to 'Ein et-Turaba (Ofer 1998: 20, 29). 'Ein el-Ghuweir was established in the

36. De Vaux 1973: 58–59, and No. 35 on the general map in de Vaux 1956: pl. XL. Patrich 1994: 74, n. 4: structure No. 35 refers to this same Iron Age structure.

37. A later cemetery with at least six tombs made of piled stones with a north–south axis, was noticed within the courtyard of the Iron Age structure (No. 11). We examined this cemetery further to the south and it appears to be quite extensive. Most of the tombs were built of small heaped stones, but there are occasionally larger more prominent tombs as well, marked with rows of stones around their edges. Most of the tombs have a north–south axis, but there are a few with an east–west axis. Scattered sherds of nineteenth–early twentieth-century jars and water jugs (including those of the “Black Gaza Ware” variety) were visible on the surface.

38. By comparison the Khirbet es-Samrah compound in the Buqei'a has an area of 0.26 ha.

39. De Vaux 1973: 58–60; 1953: 540–61. This wall was so impressive that it figured prominently on a map of the region made in 1850/51: de Saulcy 1853: second map labeled “Itinéraire du pourtour de la mer morte.”

40. De Vaux 1973: 59. A similar wall was found also running 600 m between 'Ein el-Ghuweir and 'Ein et-Turaba; see Blake 1966: 595; Bar Adon 1970: 399.

41. Bar Adon 1989, possibly related to a nearby village and caves; Bar Adon 1977: 126; Ofer 1998.



Figure 18. The northern angle of the Iron Age building at Khirbet Irneh toward the south. Photograph by S. Gibson.

late Iron Age, reestablished in the Hasmonean period, and continued to be used in Early Roman times. At 'Ein et-Turaba the same chronological parameters appear at a similar altitude (Ofer 1998: 41–49). 'Ein et-Turaba was a large well built structure with a courtyard in front, possibly dating to the eighth century BCE. At Rujm esh-Shajra, there was a late Iron Age fortified settlement near the coast (Bar Adon 1989: 86).

'Ein Gedi (Tel Goren) was undoubtedly a very important town along the western shore of the Dead Sea, thriving particularly in the seventh century BCE (Period V), with a perfume industry at the oasis.⁴²

In addition to these sites with remains of built structures, Iron Age II material, dated to the eighth to early sixth century BCE, has been found in caves in the vicinity of Qumran (Burdajewicz 2016). The 1952 survey of caves close to Qumran uncovered such material in GQ6, 13, 27, 39, "Cave A" and "Cave B" (de Vaux 1962). A small fragment from the side of a bowl was found in Cave 9Q, one of the marl caves below the end of the esplanade (de Vaux 1961: 31). In 11Q jar fragments, a rounded juglet and two Iron Age II lamps were found (de Vaux 1956: 574; 1973: 51; Humbert and Chambon 1994: 265–66 and 344). Józef Milik (Milik 1959: 151; see Taylor 2016) excavated a cave with a protruding terrace above 'Ein Feshkha, (wrongly) identified as GQ37 in

42. Mazar, Dothan, and Dunayevsky 1966; Stern 2007: 23–27. At the top of the 'Ein Gedi ascent leading from Tel Goren westwards toward Tekoa is an Iron Age II fortress; Meshel and Ofer 2008; Porat and Davidovitch 2008: 73 map of road fig. 1.

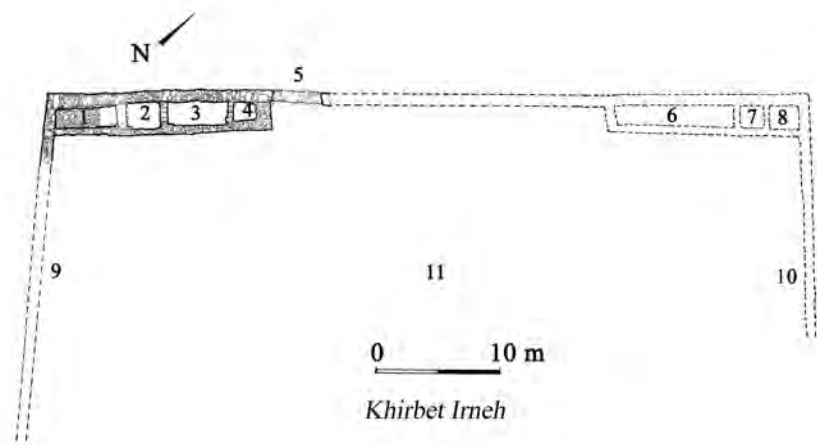


Figure 19. A plan of the range of 'casemate' rooms along the northwestern edge of the site at Khirbet Irneh, adapted from a drawing published by Humbert and Chambon (1994, pl. XLVIII), but with corrections and additions based on new measurements made by the authors in Nov 2010. Plan by S. Gibson.

Ziegler's map, with a stratum of Iron Age II sherds. Joseph Patrich's excavations of Cave 13 (probably cave GQ 2 in the 1952 survey)⁴³ also revealed Iron Age II pottery as well as a mud and palm-frond floor.⁴⁴ Patrich found the true GQ37, further north than on Ziegler's plan, also with a terrace and a mud and palm-frond floor, and containing a great deal of Iron Age II material, including two burials, one a child. There were also Iron Age II wooden beams, possibly from a ploughshare, and a juglet.⁴⁵ The University of California Long Beach found Iron Age II pottery in their survey at a site they designated as Cave 11, near the Kidron (Eisenman 2000: 126). The Operation Scroll Survey, which ranged over a large zone northwest of the Dead Sea, found 16 caves with Iron Age remains, nine of which are in the wider area around Qumran (Burdajewicz 2016: 248).

Further south, in caves 1, 2, and 3 in Wadi Murabba'at (Nahal Darga), there was pottery and other items from Iron Age II, including a letter dated to the eighth century BCE (Mur 17; Harding 1952; Benoit, Milik, and de Vaux 1961). Pessah Bar Adon identified Iron Age II sherds in caves west of Ain et-Turaba and Ain el-Ghuweir (Bar

43. Burdajewicz 2016: 247 wonders if Cave 13 was GQ37 as described in DJD 3, but actually this was another cave.

44. Patrich and Arubas 1989; see also Patrich 1994: 92 on the problem of dating this and other floors in caves.

45. Eisenman 2000: 123; Patrich 1989: 34. A decorated arrow shaft was originally thought to be Iron Age but it is now understood to be Roman, thanks to the reclassification by Guy Stiebel, see Taylor 2016: 20.

Adon 1989: 41–49). In the caves of Nahal Hever and Nahal David, close to ‘Ein Gedi, several contain Iron Age II pottery (Amit and Eshel 1998), notably the Cave of the Pool (Avigad 1961).

Mariusz Burdajewicz has suggested from the pottery that most of these caves is from the eighth to sixth centuries and in some of the caves this can be narrowed to late seventh to early sixth centuries. He suggests most caves would have served as temporary shelters, for seasonal occupation, or as hideaways (Burdajewicz 2016: 253). One may note though that there is a burial in GQ37. There is some question about the marl caves at the end of the Qumran esplanade, overlooking the wadi. In an interview with Hershel Shanks, Frank Moore Cross suggested that the marl caves (4Qa, 4Qb, 5Q, 7Q–9Q) might be former Iron Age tombs.⁴⁶ Cross’s suggestion may be supported somewhat by the fact that at Megiddo and Tell es-Saideyeh there are tombs cut into the edge of the Iron Age tells themselves, close to the settlements, and other hewn cave tombs of the Iron Age occur on the edge of wadis (Bloch-Smith 1992: 20, 39, figs. 17 and 18, cf. 256–57). Elizabeth Bloch-Smith notes that cave tombs are found in the “soft chalk and limestone outcrops of the foothills and highlands east and west of the Jordan River” and that such tombs were “initiated during the tenth through the sixth century BCE at the eastern Cisjordanian foothill and highland sites of Khirbet Beit Lei, Bethlehem, Lachish, Tell Judeidah, Ras-et-Tawil, Khirbet Rapud and Ez Zahariyah, and at Amman and Mt Nebo in Transjordan” (Bloch-Smith 1992: 39). Such tombs “cluster on the slopes below or outside a settlement, in rock cliffs facing a settlement” (Bloch-Smith 2013: 254).

Alternatively, as Burdajewicz suggests, it is possible that the marl caves do date from the Iron Age, but they were used for habitation. Burdajewicz (2016: 254) notes importantly that the majority of caves used in Iron Age II lie close to Qumran: “It is a marked cluster of caves in close proximity to the settlement on the plateau” (and see his fig. 17.1), and thus they deserve to be called “Qumran caves.”

Cross-Temporal Study: Qumran in the Iron Age and in the Hasmonean Era

Having reviewed the evidence for Qumran in the Iron Age II, and considering also the contemporaneous environment, it appears that the settlement at the site was much more extensive than has previously been supposed, even by de Vaux’s estimation, which is perhaps the reason why he decided not to define the cultural material from this time frame as an independent “period” in his overall chronological scheme of the site. Indeed it is surprising, but scholars continue to ignore or minimize the existence

46. This identification is not widely known, and we are grateful to Sidnie White Crawford for pointing this out to one of the authors (Taylor); Shanks 1994: 114. Alternatively, Magen and Peleg (2018: 100) surmise that the cemetery of trench-dug graves on the east side of Qumran had its beginnings in the Iron Age but they offer no evidence to support this idea.

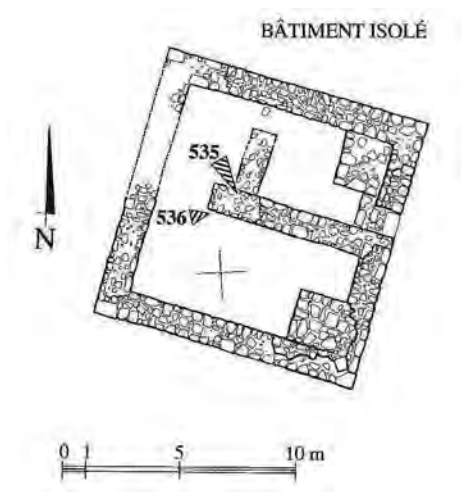


Figure 20. A plan of the Iron Age “bâtiment isolé.” Adapted from Humbert and Chambon 1994, pl. XLVIII.

of this significant pre-Hasmonean stratum at the site, especially given that it has largely not been excavated down to this level.⁴⁷ What has been probed suggests a substantial settlement. This oversight is largely due to the great interest in periods that relate to the Dead Sea Scrolls, rather than in earlier centuries. This is unfortunate because, as we shall see, the overall architectural layout of the Iron Age settlement had a significant impact on the eventual appearance of the early phase of the Period Ib Late Hellenistic/Hasmonean building complex at the site (fig. 7).

Analysis of the architectural remains dating from the Iron Age suggests an overall rectangular building complex (33 × 36 m), roughly resembling in character known building plans of that period elsewhere,⁴⁸ with stone-built foundations and a superstructure built of mud bricks (fig. 21). It comprised a tower-like domestic area protruding at the northeast angle of the compound (10 × 10 m), an industrial area in a corridor (10 m wide) within the southeast angle, probably open to the sky, another two corridors (10 m wide) to the south and west (L.4/1), which may have been subdivided originally into rooms, a very narrow corridor (L.38) on the north (6 m wide), and a large open square courtyard at the center of the complex (16 × 16 m), with open-air

47. Magen and Peleg 2018: 101–4 perpetuate this by not including a chapter on the stratigraphy of the Iron Age remains at Qumran in their final excavation report, except for a summary of the period in a chapter on the history of the site. Magen and Peleg 2018: 11, 103 sum up the Iron Age remains at Qumran as reflecting a temporary civilian and/or seminomadic settlement used on a seasonal basis in the winter months, with flimsy structures and tents.

48. For example, the Iron Age IIC site of Khirbet Abu et-Twein: Mazar 1982: 174. It had a rectangular plan with rooms surrounding a courtyard, and some of the surrounding long rooms were partitioned with rows of stone pillars. This might suggest that we should identify the north–south partition wall in L.51 at Qumran not as a full constructional wall but as the stone foundation for a row of monolithic stone pillars, such as can be seen at many Iron Age II sites in Judah.

working surfaces situated further to the south (e.g., the plastered floor beneath L.77). In addition, there was a smaller self-contained building on the west (19×15 m) which had a round cistern, conduits, and adjacent installations. The western building was separated from the main building by an alley in L.107 (8 m wide). Various add-on buildings probably existed as well, to the east and south, and a long wall ran along the eastern edge of the southern esplanade. Altogether, the built-up area covered an area of only 1.5 dunam, or 0.15 ha. The domestic area was limited, and most of the space was given over to water-harvesting, industrial activities, and storage. It is difficult to suggest a figure for the number of inhabitants living at the site in the Iron Age II because some of them may have been living in tents on the esplanade to the south,⁴⁹ but it is reasonable to assume that it was no more than two to three extended families.

The main dating evidence for the Iron Age occupation at the site is based on the pottery derived from de Vaux's excavations, which remains unpublished, but the overall assemblage of common wares is attributable to the seventh–sixth centuries BCE.⁵⁰

While the Period I building complex at Qumran was evidently constructed sometime in the Late Hellenistic/Hasmonean era, various opinions have been given in regard to the date of its initial phase (Period Ia), with a starting date ranging between 150 and 100 BCE.⁵¹ Our impression of the few intact artifacts that might be attributed to Period Ia, notably an intact cooking pot from under an oven in L.30, and a few bowls, is that they belong to ceramic types generally attributable to the last third of the second century BCE, but equally they could date from the very end of this time range as well, that is, to ca. 100 BCE, and thus to the very beginning of Period Ib as defined by de Vaux. This is also true of the late Hellenistic range of lamps from the site (Types 031, 032–035, 041–044), which conceivably might be of late second-century BCE date but are more likely to be from ca. 100 BCE at the earliest (Młynarczyk 2013). Importantly, the Period Ib pottery at Qumran is identical to Hasmonean pottery from Jericho, which is attributed to no earlier than 90 BCE (Bar-Nathan 2006). One should also mention the evidence of Hellenistic coinage (Ptolemaic and Seleucid) indicating a general date in the last quarter of the second century BCE, which might be significant, but one has to take into account the possibility that some of these coins, especially the silver ones, remained in circulation until very late in that century and beyond.⁵² The foundation

49. On the evidence for the use of tents at Qumran, albeit in later periods, see Homan 2002: 54–55.

50. One cannot deny that the settlement at Qumran might have been established already in the late eighth century BCE, but we think this unlikely. In the conclusion to Yezerski's study of the Iron Age pottery from the Magen/Peleg excavations, she suggests a mid-seventh century to early sixth century BCE date for the assemblage, though how this fits with the evidence of the two *lmlk* stamped handles from the site is left unresolved; Yezerski 2018: 347.

51. Davies 1982: 54; Cargill 2011. For a summary of the various views see Mizzi 2015.

52. De Vaux 1973: 18–19; and for the coins from L.120: Humbert and Chambon 1994: 329–30. A tetradrachm of Demetrius II from 129 BCE has recently come to light in the southern esplanade: Farhi

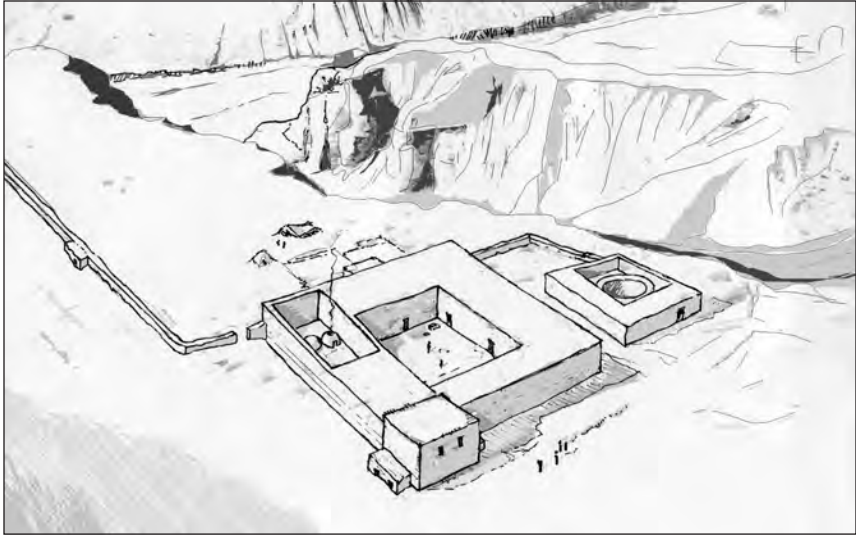


Figure 21. A reconstruction drawing of the Iron Age settlement toward the southwest with Wadi Qumran behind. Drawing by S. Gibson.



Figure 22. Levelled areas immediately to the north of Qumran which were probably adapted in antiquity for green pasturage. Note the remnants of a terrace wall in the foreground. Photograph by S. Gibson.



Figure 23. The authors, Taylor and Gibson, in Pool L.49, deliberating the architectural and stratigraphic phasing of the site at Qumran. Photograph by D. Gibson.

date in ca. 100 BCE places the Hasmonean settlement of Qumran in the reign of Alexander Jannaeus (103–76 BCE).⁵³

This suggests that de Vaux's subsequent Period Ia in the second century BCE did not in fact exist, as originally suggested by Magness (2002: 64–68; Mizzi and Magness 2016: 301–2). It consists of fills with an amalgamation of mixed cultural materials derived from two strata: residual items from the earlier Iron Age level, on the one hand, and intrusive items from the very first phase of the Late Hellenistic/Hasmonean era (Period Ib, i.e., from no earlier than ca. 100 BCE), on the other. As we have seen, quite a few Iron Age stratigraphical features were incorporated erroneously by de Vaux into his Period Ia. Added to this, we have the new Iron Age material uncovered by Magen and Peleg that confirms the importance of the Iron Age stratum at Qumran, despite their own interpretations. There is now a growing understanding that the chronological scheme of periods employed by de Vaux at Qumran no longer works, and that his Period Ia is actually a conflation of a number of different phases of construction (Mizzi 2015; Mizzi and Magness 2016).

While the first cohesive Hasmonean settlement at Qumran appears to have been established no earlier than the very beginning of the first century BCE, with the Period Ia phase of the late second century BCE now quite dubious, it is clear that there was an initial building phase in which there was a Hasmonean square enclosure, but floors have not survived in situ (except for traces of an oven in L.30). A few residual Late Hellenistic pottery types are in mixed contexts from Period Ib. If we retain de Vaux's terminology, for the purpose of the present paper, we need at least to see Period Ib as a phase that had two stages of development: (1) an initial Hasmonean square enclosure and (2) an expanded form with complex water systems and industrial units.⁵⁴ For the overall layout of the Hasmonean phase of Qumran, we follow what has been presented previously in the work of Humbert.⁵⁵

But how does our reconstruction of the layout of the Iron Age settlement compare with the layout of the subsequent Hasmonean (Period I) settlement and its Roman successor?

Without going into any minor issues, the overall points of similarity are:

1. *A need for a viable and sophisticated water collection system.* In the Iron Age, the harsh environment made water harvesting an imperative at Qumran. The "cistern

and Price 2010. See now seven additional Ptolemaic and Seleucid coins from excavations at Qumran in Ariel 2018.

53. For further on the historical background and dating of what should just be called Period I, see Taylor 2012: 225, 227, 251–52.

54. The form of this initial Hasmonean construction has been explored most extensively in Humbert and Chambon 1994: 419–44, 467–82.

55. We are grateful to Jean-Baptiste Humbert for sending us a forthcoming paper that summarizes his arguments to be published in the next Qumran volume, and focuses on comparanda of other square structures.

house,” on the west side of the settlement, functioned sufficiently well for it to be serviced by two direct access stepped reservoirs. The pottery-making procedures would have required a variety of features and water was of course essential for other purposes as well. During the Hasmonean era this water-collection system was expanded, but initially, during the early phase of Period I, the original Iron Age water system was simply cleaned, repaired and put back into use. The continuity in the use of water installations spanning different periods is a well known phenomenon at Iron Age to Roman sites in the southern Levant.

2. *Pottery-making.* Pottery making is a feature of both the Iron Age and the Hasmonean-Roman periods (Periods I–II). There appears to have been two kilns functioning simultaneously in the Iron Age, accessed from different directions. Petrographic studies should now be made on the Iron Age pottery from Qumran to see if it is possible to ascertain which of the pottery vessels were made locally and which types were brought in from elsewhere.⁵⁶ In the Hasmonean-Roman era there were also two kilns, one kiln in L.64 for making larger ceramic vessels and one for smaller vessels, also accessed from different directions just as they were in the Iron Age. Pottery-making was undoubtedly a key activity in Iron Age Qumran, but not necessarily the only one. Our lack of information concerning the nonceramic finds from the Iron Age levels at the site (e.g., animal bones, carbonized seeds, fibres, etc.), does not mean we should just assume the settlement at that time was only a pottery-making facility. Other multiresourcing activities may have included: the gathering of salt and bitumen from the Dead Sea for trade, weaving and rug-making, the up-keep of areas for green pasturage to the north and south of the settlement on leveled and terracing land,⁵⁷ and the herding of livestock (sheep/goats) and exploitation of their secondary products (milk, wool, and hair; fig. 22).

56. Yezerksi 2018: 339, but she does not dwell much on fabrics. The vessels were moderately well-fired, with slips, some of the red variety, and many ring burnished. The ware contained varying quantities of large and small white limestone grits. The percentages of nonslipped vessels and burnished vessels were not calculated. Yezerksi does, however, indicate that similarities of material in terms of techniques of production and surface treatment imply that some vessels were made in a “shared workshop” (p. 340). In his petrographic study of the Iron Age II pottery assemblage from the Buqe‘a sites on the plateau west of Qumran, Master 2009: 308 notes that “a surprising amount of the domestic pottery was made locally.”

57. Clearly, straightforward agricultural pursuits would not have been possible in the immediate Qumran environment during the Iron Age, owing to the constraints of a harsh arid climate and a lack of water, notwithstanding the fact that the climate at that time was a little better than it is today (see below, n. 62). Later, during Hasmonean/Roman times, the growing of date trees was a major commercial venture practiced in the region, and these plantations required high maintenance. We believe a few date trees were probably grown on the southern esplanade of Qumran, and the press unearthed at L.75 was probably used for making date wine (contra Magen and Peleg 2018: 90 and fig. 108, who argue that it was used for water sedimentation). In addition, a date palm grove must have existed at ‘Ein Feshka; Broshi 2006.

3. *Orientation.* The orientation of the Iron Age structures is replicated in the later Hasmonean-Roman constructions. As was common, a ruin provided readily available building materials and a template for the layout of later buildings. Fallen walls were raised and new walls were built on old foundations. This resulted in the “disappearance” of Iron Age walls beneath the lower parts of Hasmonean walls. Additional Iron Age walls were likely to have been quarried for building stones, during the sinking of foundations in later periods, which further resulted in the diminishing of the preservation of the earliest remains at the site. It is not surprising therefore that the oldest remains are also the worst preserved.

4. *Burning.* Given the extent of the ubiquitous ashy layer, it is likely that the Iron Age settlement was destroyed by a devastating fire caused by an invading army in 587/586 or 582 BCE. Another fire from the invading army of the Romans devastated the site in 68 CE. In between, there may have been occasional localized fires (see Wagemakers and Taylor 2011).

Differences

1. *Size.* Overall, it seems the size of the occupied space of the main enclosure in the two periods is not considerably different. However, as Period Ib developed, the western sector of the site was expanded and divided up into much smaller interior spaces, and the overall layout became more complex, with the addition of a very large defensive tower on the north. The water systems were much expanded and developed with many more pools. The need for expanded water systems, given the relatively small size of the site, is striking, but it does not seem to be a direct result of the clay-preparation procedures as Magen and Peleg have suggested. The extension of the water system might indicate an increasing need for water in the Hasmonean-Roman periods for the special activities of the site, with a greater reliance on capturing the flood waters from the wadi to the west, with water carried down by aqueduct.⁵⁸ By comparison the Iron Age water system

58. Ilan and Amit 2002. The original date of the aqueduct is uncertain, but it was undoubtedly in use during Period II; de Vaux 1973: 8 attributed it to Period Ib, but Humbert in Humbert, Chambon, and Mlynarczyk 2016: 425–32 now suggests it was established after ca. 31 BCE. In November 2009 a trip was organized by one of the authors of this paper (Gibson) to examine the outlet at the eastern end of the aqueduct, with the participation of the late David Amit, Stephen Pfann, the late David Stacey, and the late Yuval Peleg. We examined the following locations: (1) the base of the “waterfall”; (2) the supposed barrage/dam and its axis, and the area behind where water might have been intentionally “ponded”; (3) the rock-cut aqueduct; (4) the tunnel and its relationship to a segment of looped aqueduct; and (5) the possible rock-cut spillways to the south. Peleg was of the opinion that the barrage/dam never existed, and that the aqueduct only collected water from the gullies leading from the southern side of the wadi, that the “loop” was a first and unsuccessful attempt to create the aqueduct, and that it was eventually superseded by the tunnel. He also thought the spillways (one or all three of them) are natural. Peleg indicated that he had prepared a new map of the aqueduct, which he planned to update as a result of our November field trip. David Amit disagreed with Peleg in regard

seems quite small, and was entirely dependent on rainwater and surface flooding from the immediate area round about.

2. *Complexity.* Along with the size of the settlement, the later settlement has a much more complex layout, indicating it had numerous new functions, though this assessment must take into account that much of the Iron Age architectural remains are either unexcavated or in a bad state of preservation. This becomes increasingly apparent as the site develops through the first century BCE and first century CE, with a nonsymmetrical pattern which deviates from the symmetry of both the Iron Age and initial Hasmonean phase building complexes. The nonsymmetrical and expansive developments include a meandering water system, the industrial complex on the west, and the extensive use of marl and limestone caves in the vicinity of the site (Broshi and Eshel 2004).

3. *Continuity.* While the Iron Age site did not recover after its destruction in the sixth century BCE, the site of Qumran in Period II did not suffer the same devastation that made reuse of it impossible, and it continued to be occupied in Period III, at least until the 90s of the first century CE and finally in the Bar Kokhba period (Taylor 2006; 2012: 261–65).

4. *Relationship.* While in the Iron Age it is easy to see the close relationship Qumran had with its many surrounding settlements, to which it was connected via a network of roads and paths, the relationship between Qumran in the Hasmonean–Roman period and other settlements is less than clear. In other words, Qumran of the Hasmonean–Roman period was apparently much more isolated in its regional setting than in the

to the looping aqueduct being an abandoned attempt at creating an aqueduct, and pointed out that plaster was visible adhering to the side wall of this aqueduct, at the point where it linked up to the tunnel, and therefore the plaster must indicate that it had functioned at some point. Amit also believed that at least one of the spillways to the south was artificially cut, notably the one leading from the wadi to the southeast, which he said was much better preserved in tunnel form and with tooling seen on its sides when he first investigated it more than 20 years earlier (this feature is No. 5 on his published map), extending to a deep cut in the rock scarp which is clearly an outlet for surplus water dropping down to the wadi below. The existence of a barrage/dam does fit the topography and would have been a logical feature, so as to collect the maximum of water especially at times when the flow of water was at a minimum. Hence, water ponding behind the barrage wall would have reached a height of at least 2 m. Nothing is now visible of any construction or wall, but this is probably because it was washed away by flash floods which in the winter descend in this area with a ferocious energy. The spillway channel (No. 5) would have taken surplus waters away to the southeast from behind the dam when flooding occurred. Moreover, on the opposite side, there is a level area which may represent the remnants of a channel linking the area behind the barrage to the v-junction to the east of the aqueduct and its southern loop. All the participants agreed that the construction of this aqueduct would have required substantial engineering expertise and financial investment. Compare these comments to those in Stacey and Doudna 2013. Magness 2014 disagrees with Stacey's date for the original construction of the aqueduct in Period II, and follows de Vaux's dating for it in the Hasmonean Period Ib. The Hasmonean date is also maintained by Magen and Peleg 2018: 75–80, figs. 86–88 who compare it to the Hasmonean aqueduct known from nearby Jericho and thus date it accordingly.

previous Iron Age.⁵⁹ The nearby site of 'Ein Feshkha was developed in the Hasmonean period, as were other Iron Age settlements to the south (Kh. Mazin, 'Ein el-Ghuweir) and to the north (Rujm el-Bahr), but these by comparison are smaller and much further apart in distance than the corresponding Iron Age settlements (e.g., the short distance between Qumran and Kh. Irneh). There are no settlements in the Buqei'a until later in the Roman period, perhaps corresponding to Period III in Qumran. Qumran therefore appears much more isolated in these later periods (fig. 23). Pastoralist groups in this region may have been impoverished, or intentionally excluded from the region owing to the development of date plantations and with the build-up along the coast of sea-faring activities for the gathering of bitumen, sulphur, and salt.

Conclusions

The Iron Age building complex at Qumran fits very well the picture of expanding and burgeoning settlements along the eastern border of the Kingdom of Judah in the seventh century BCE.⁶⁰ It is during this century that the central state authority was actively strengthening its eastern defences against possible pillaging and encroachment of military incursions from Transjordan (Ammonites and/or Moabites), by building very small fortified outposts, by developing networks of roads/paths, by encouraging economic advantages, such as the extraction of salt and bitumen from the Dead Sea, by providing safe passage for caravans of traders crossing through this area, by instigating industrial and agricultural activities, and by supporting the needs of local nomadic and semipastoralist groups in order to gain their loyalty (Stern 1994).

Hence, we argue against the notion that the catalyst for the flourishing of settlements along the eastern border of Judah was the direct result of an influx of refugees into the Kingdom of Judah from the north, following Sennacherib's successful invasion of Samaria in 720 BCE, who were forced by circumstances to adopt a seminomadic existence (Magen and Peleg 2018: 101–4; 2006: 73–79), on the grounds that at times of major stress and hardship refugees from towns/cities head toward urban centers and not into marginal environments, such as the Judean Desert, which had limited water sources. The oasis of Jericho would be a good destination, but not the Buqei'a plateau. In general terms, the settlement process in this region was most likely the result of a state initiative concerned with protecting the eastern border of Judah against military incursions from the east. This was not done by erecting large numbers of forts, but by “filling the vacuum” by establishing a plethora of very small enclosed *ḥsr* (חצר) settlements with different functions,⁶¹ and by securing the allegiance of nomadic, semisedentary and pastoralist groups in the area.

59. For an alternative approach, see Zangenberg 2004.

60. On the economic prosperity of the southern Levant in the seventh century BCE under Assyrian control, see Faust and Weiss 2005.

61. The *ḥsr* (חצר) could equally refer to a small enclosed agricultural farmstead in highland

Technological know-how might also have been provided by the Judahite state in facilitating the establishment of water collection and harvesting systems at sites, as can be seen at Qumran and perhaps also in the Buqe'a, which would have encouraged people to settle in such marginal arid environments.⁶² In addition, there can be no doubt that the road and path systems of the region were substantially improved and developed at this time, and this was unprecedented, as we were previously able to show through our work in the region of Qumran (Taylor and Gibson 2011). The opening up of these roads and paths to human traffic, clearing them of stones and maintaining them from winter flooding, was not done on an ad hoc basis; protecting them was imperative for the safety and economy of the Kingdom of Judah in the seventh century BCE and it provided considerable political and financial advantage.

The Hasmonean settlement at Qumran (beginning ca. 100 BCE) was far more isolated than its Iron Age predecessor, and it was connected to only a few local sites along the Dead Sea. The Hasmonean complex had a defensive tower, but was otherwise built very much on the same template as the preceding Iron Age structure, adopting the same kind of symmetrical plan and utilizing surviving walls, to such an extent that it is sometimes impossible to separate out the Iron Age and Hasmonean phasing in individual walls. In addition to this problem, the Period I settlement at Qumran undoubtedly had at least two construction stages during the course of the first half of the first century BCE, and distinguishing between the architectural features of the two is complicated. The basic plan of the initial Hasmonean-period resettlement, which may be closely associated with the early building works at Rujm el-Bahr and Khirbet Mazin/Qasr el-Yahoud, was eventually replaced with a far more-developed water system, an industrial center, cave use, and a complex of buildings that deviates from the original symmetrical plan. While these new developments may be the direct result of the economic advantages of maintaining large date plantations in the area, they also indicate that the site of Qumran, in terms of its layout and functions, had now become

environments (e.g., in the region of Jerusalem), or to an enclosed small settlement used by semi-sedentary pastoralists (e.g., in the Judean Desert), as one of the present authors already pointed out long ago, based on archaeological surveys of Iron Age sites around Jerusalem: Edelstein, Gat, and Gibson 1983 (Hebrew). See now a comprehensive discussion on the terms *בית החצר* and *חצר* in Faust 2008a: 109. On settlements in the Iron Age II with clearly defined boundary walls or fences, see Faust 2017.

62. Notwithstanding the current low rainfall in the Dead Sea region, it would appear the climate of the southern Levant was wetter in the Iron Age. However, this does not mean that the Qumran environment in the Iron Age was markedly different from today's. One must take into consideration the variable climatic fluctuations that existed at that time, and the impact these might have had on human populations at the micro level, especially to those vulnerable to drought, which can occur even in regions with temperate climates. Indeed, very small shifts in climate in an arid zone can be disastrous for the survival of pastoralist groups and their flocks. On the evidence for the climate in the Iron Age, see Neumann et al. 2010: 760–62; and the summary in Langgut et al. 2015: 230.

fundamentally distinct from the much older Iron Age and earlier Hasmonean building complexes that preceded it.

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